

Agent Exploration Toward Artificial General Intelligence: A Survey

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Abstract

Exploration is not an optional behavior in natural intelligence; it is an evolutionary principle underlying the emergence and adaptation of intelligence. Curiosity and deliberate probing emerge as evolved responses to uncertainty,

enabling organisms to expand competence, and preserve adaptability in changing environments. In this survey, we argue that this evolutionary logic is equally indispensable for artificial general intelligence (AGI): exploration is not a heuristic appended to an agent, but the mechanism through which generality becomes possible. It is not enough for AI systems to learn only what they are expected to learn. To move toward AGI, agents must also know what they do not know and actively explore. We propose a unified view of *Epistemic Exploration* for agentic systems: the capacity of an agent to actively acquire information that reduces uncertainty about the world, seek experiences at the boundary of its current capabilities, and preserve epistemic reachability as the readiness and ability to adapt when the world changes. This view yields three criteria: Information Gain, Value Improvement, and Epistemic Reachability. We then introduce an exploration-centered five-level trajectory toward AGI and organize related works along this trajectory. In this framework, each level is defined by a distinct exploration capacity: (1) *Responder*, with minimal or no explicit epistemic exploration; (2) *Reasoner*, with reasoning-space exploration for hypothesis search, branching, backtracking, and self-verification; (3) *Agent*, with interaction-space exploration through tool use, memory, embodied perception, and closed-loop action; (4) *Prospector*, with imagination-space exploration over learned world models and counterfactual futures; and (5) *Ecosystem*, with coordination-space exploration among heterogeneous agents. Finally, we conclude with evaluation principles and open challenges for building exploration-centric agents. By viewing exploration as the common thread underlying modern AI systems, this review provides a unified perspective that connects LLM reasoning, agent harness, embodied AI, world models, and multi-agent intelligence within a single conceptual framework.

Keywords: Agent Exploration, Epistemic Exploration, Artificial General Intelligence, Reinforcement Learning, LLM Reasoning, Embodied AI, World Models, Multi-Agent Systems

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1 Introduction

Exploration is not an optional behavior in natural intelligence; it is an evolutionary principle underlying the emergence and adaptation of intelligence. From infants who

THE FIVE LEVELS TOWARD AGI

Exploration as the Transition Mechanism

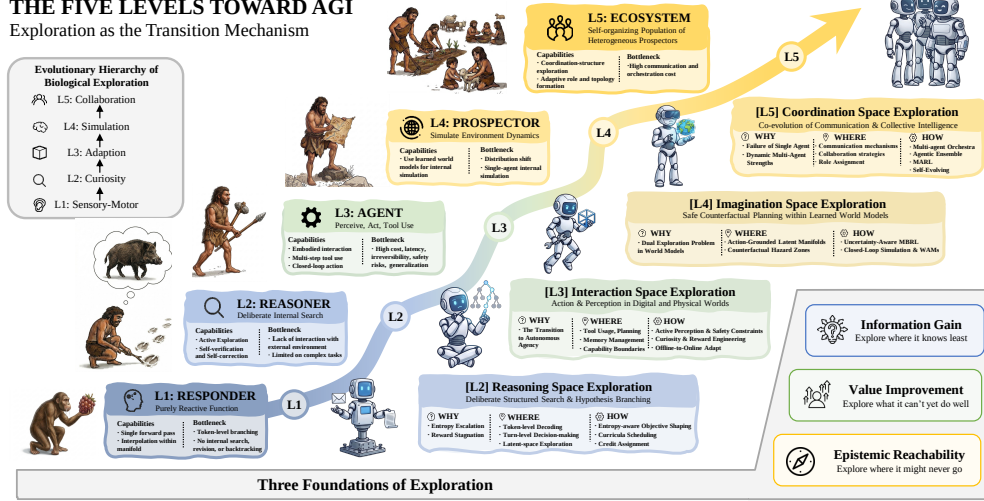


Fig. 1: Five-level trajectory toward AGI through Epistemic Exploration. The framework views exploration as the transition mechanism from *Responder* to *Reasoner*, *Agent*, *Prospector*, and *Ecosystem*. Across these transitions, agents progressively expand exploration from reasoning space to interaction space, imagination space, and coordination space, guided by three foundations: information gain, value improvement, and epistemic reachability.

attend to surprising events to animals that probe novel regions even when immediate exploitation is available, structured information seeking appears before explicit instruction [1]. Organisms explore not as an incidental byproduct of behavior, but as an adaptive strategy for reducing uncertainty, discovering causal regularities, and expanding the repertoire of skills they can reliably deploy.

Large language models (LLMs) have achieved striking success in relatively closed-world domains, including mathematical reasoning, code generation, and standardized benchmarks. Techniques such as chain-of-thought prompting, problem decomposition, and inference-time compute scaling have substantially improved the ability of models to reason within fixed contexts. Yet these advances largely operate within short-horizon, static inference regimes: they enhance how a model solves a given problem, but do not by themselves equip it to actively seek missing information, adapt to changing environments, or improve through sustained interaction with an evolving world. It is not enough for AI systems to learn only what they are expected to learn. To move toward AGI, agents must also know what they do not know—and actively explore to acquire new information, capabilities, and experiences.

In this survey, we systematise this challenge through the framework of **Epistemic Exploration**. Our central claim is that general intelligence cannot be achieved through passive sequence generation, reactive input-output mapping, or static optimisation alone. An intelligent agent must act with an epistemic exploration: it must acquire information when its beliefs are incomplete, seek experiences near the boundary of its current capabilities, and preserve access to alternative states, strategies,

and information sources so that future adaptation remains possible. These requirements correspond to three core criteria: *information gain*, *value improvement*, and *epistemic reachability*. Together, they provide a unified lens for understanding how reasoning, action, world modelling, and coordination emerge from the same underlying imperative: to reduce uncertainty, expand competence, and keep future learning possible.

Definition of Epistemic Exploration

Epistemic exploration is the capacity of an agent to actively acquire information that reduces uncertainty about the world, seek experiences at the boundary of its current capabilities and convert them into durable capability improvement, and preserve epistemic reachability as the readiness and ability to adapt when the world changes.

This definition leads to two complementary axes for organising the survey. The first is a *foundational axis*, which asks why exploration is necessary, what it must accomplish, and how it is realised in practice. Static optimisation fails in open environments for three structural reasons. First, *belief stagnation* occurs when the agent’s model of the world becomes stale under distributional shift, yet the agent lacks a mechanism to detect or correct that drift. This motivates Information Gain: the agent must actively seek observations that reduce epistemic uncertainty. Second, *value stagnation* occurs when the agent either fails to encounter experiences that challenge its current competence or fails to convert new information into improved future behaviour. This motivates Value Improvement: exploration must target the boundary where learning can produce durable capability gains. Third, *reachability collapse* occurs when the agent’s behaviour contracts prematurely onto a narrow region of the state, action, or strategy space, thereby destroying the conditions required for future learning. This motivates Epistemic Reachability: the agent must preserve access to plausible alternatives before adaptation becomes impossible.

These three criteria also correspond to three recurring methodological families. *Uncertainty-driven exploration* primarily serves information gain: the agent explores where its predictions are least reliable, using signals such as entropy, ensemble disagreement, predictive variance, or confidence margins [2]. *Competence-driven exploration* primarily serves value improvement: the agent targets the frontier of its current abilities through curricula, self-play, failure-conditioned sampling, and adaptive task selection [3]. *Reachability-driven exploration* primarily serves epistemic reachability: the agent preserves behavioural diversity and future options through coverage bonuses, entropy regularisation, anti-collapse constraints, and anti-foreclosure mechanisms[4]. These families are complementary: uncertainty identifies where information is missing, competence signals identify where learning can improve the agent, and reachability mechanisms ensure that future learning remains possible.

The second organisational axis is a *trajectory axis*, which traces how exploration expands across five levels on the path toward AGI. Existing taxonomies of AGI progress, including OpenAI’s five-level framework [5], primarily organise systems by

the capabilities they exhibit. Our framing is complementary but asks a different ques-	185
tion: not only <i>what</i> a system can do at a given level, but <i>what kind of exploration</i>	186
it possesses. We argue that each transition toward more general intelligence requires	187
opening a qualitatively new exploration space.	188
Level 1: Responder. A Level-1 <i>Responder</i> produces outputs through learned	189
input–output mappings, often in a single forward pass. Such systems may be flu-	190
ent and useful, but their exploration capacity is minimal: they do not deliberately	191
search hypotheses, revise intermediate conclusions, or verify their own reasoning.	192
Their behaviour is therefore largely confined to reactive generation within the learned	193
distribution.	194
Level 2: Reasoner. A Level-2 <i>Reasoner</i> opens reasoning itself as an exploration	195
space. Instead of producing a single response directly, it can branch across reasoning	196
trajectories, evaluate hypotheses, backtrack from errors, and select paths that reduce	197
uncertainty about the correct conclusion. At this level, exploration occurs over latent	198
thoughts, token-level branches, and intermediate reasoning paths.	199
Level 3: Agent. A Level-3 <i>Agent</i> extends exploration beyond internal delibera-	200
tion into the external world. It can perceive environments, invoke tools, query external	201
resources, use memory, act under partial observability, and update beliefs from real	202
consequences. Here, exploration moves from reasoning-space search to interaction-	203
space probing: the agent obtains information that cannot be recovered through	204
internal reasoning alone.	205
Level 4: Prospector. A Level-4 <i>Prospector</i> internalises a world model and	206
explores imagined futures before acting. Because real interaction is often costly,	207
slow, risky, or irreversible, imagination-space exploration allows the agent to simu-	208
late counterfactual trajectories, evaluate long-horizon plans, and refine policies before	209
committing to real-world action. This creates a dual exploration problem: the agent	210
must explore the real environment to improve its model, while also exploring imagined	211
trajectories to improve its behaviour.	212
Level 5: Ecosystem. A Level-5 <i>Ecosystem</i> extends exploration from an individual	213
agent to a population of heterogeneous agents. A single prospector remains bounded	214
by what one agent can discover alone; coordination-space exploration enables agents	215
to communicate, divide labour, share representations, specialise roles, and co-evolve	216
collaborative strategies in open environments. At this level, intelligence becomes not	217
only an individual property, but an emergent property of a population.	218
Across the five levels, exploration progresses from minimal local variation to	219
structured reasoning-space search, interaction-space probing, imagination-space simu-	220
lation, and coordination-space co-evolution. We analyse each level through a consistent	221
structure: Why exploration is necessary, Where it takes place, and How the three	222
exploration methodologies are realised. This structure allows us to connect mathe-	223
matical reasoning, code generation, tool use, embodied robotics, autonomous driving,	224
world-model learning, scientific discovery, and multi-agent coordination under a single	225
epistemic principle. Across these domains, the central challenge is not merely how to	226
optimise a fixed objective, but how to build agents that can continually reduce uncer-	227
tainty, improve their capabilities, and preserve the possibilities required for future	228
adaptation.	229
	230

Contributions. This paper calls for placing agent exploration at the center of future AGI research. We develop a unified framework of epistemic exploration, arguing that exploration is the transition mechanism through which agents progress toward general intelligence. Our contributions are as follows:

- **Conceptual framing.** We formalise a unified view of *Epistemic Exploration* for agentic systems, establishing a three-criterion foundation—information gain, value improvement, and epistemic reachability—and linking each criterion to a distinct failure mode of static optimisation.
- **Five-level trajectory.** We propose an exploration-centered five-level trajectory toward AGI—*Responder*, *Reasoner*, *Agent*, *Prospector*, and *Ecosystem*—and use it to organize related work across different stages of agent development. In this framework, exploration serves as the transition mechanism between levels, with each level introducing a qualitatively new exploration space.
- **Applications and evaluation.** We review representative applications and evaluation principles to show how epistemic exploration mechanisms are instantiated, measured, and assessed across domains including LLM reasoning, agentic systems, embodied AI, world models, and multi-agent systems.

2 Foundations of Epistemic Exploration

This section establishes the foundations of epistemic exploration. We first motivate exploration from biological intelligence and the limits of pure exploitation, then define its three core criteria—*Information Gain*, *Value Improvement*, and *Epistemic Reachability*—within a unified objective. We finally summarise three practical methodologies: uncertainty-driven, competence-driven, and reachability-driven exploration.

2.1 Biological Motivation and the Necessity of Exploration

Before formalising epistemic exploration for artificial systems, it is useful to ask why exploration appears so pervasively in biological intelligence. Across species, exploration is not a peripheral behaviour but a primary adaptive strategy: organisms actively interact with their environments to acquire information, reduce uncertainty, expand competence, and preserve future options [6]. From foraging animals searching sparse resources to humans probing uncertain tasks, exploration reflects a general solution to the problem of living in worlds that are partially observable, non-stationary, and too complex to be fully specified in advance.

Why nature must explore.

Classical work by Berlyne [7] identified exploration as behaviour elicited by novelty, complexity, surprise, and incongruity, distinguishing perceptual from epistemic curiosity and specific from diversive exploration. Contemporary accounts reinterpret exploration functionally: it is not a single behaviour, but a family of information-acquisition strategies that improve an organism’s predictive model of the world [8]. This view places exploration within the broader explore-exploit trade-off, while

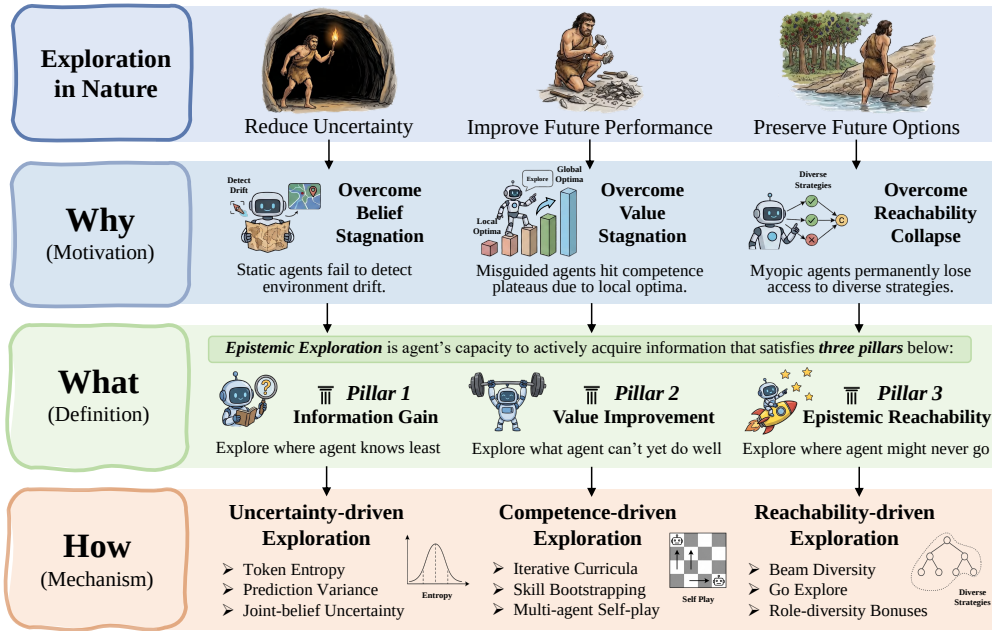


Fig. 2: Foundation of epistemic exploration. The figure connects biological exploration with the formal framework developed in this section: natural exploration reduces uncertainty, improves future performance, and preserves future options; static agents fail through belief stagnation, value stagnation, and reachability collapse; and epistemic exploration addresses these failures through three criteria—information gain, value improvement, and epistemic reachability—implemented by uncertainty-driven, competence-driven, and reachability-driven mechanisms.

emphasizing that animals often seek information even when it yields no immediate tangible reward. Exploration is therefore not merely a byproduct of reward maximisation; it is itself a computational objective.

Exploration is structurally necessary in biological systems for three reasons. First, environments change. Predator–prey co-evolution, seasonal variation, and diminishing resources make fixed strategies progressively suboptimal. Because organisms only observe the world through limited sensory histories, they must continually sample the environment to prevent their internal models from desynchronising from reality. Second, organisms must improve, not merely survive. Curiosity and intrinsic motivation direct behaviour toward situations that produce learning progress, allowing biological agents to construct self-organised curricula that expand their competence over time. Third, organisms face hard bottlenecks in energy, time, and cognition. Without mechanisms that maintain broad access, their reachable state space contracts: unexplored regions remain unknown, alternative strategies are never tested, and behaviour becomes fragile under environmental change. Collective exploration partially overcomes this limitation by distributing risk and information across a population, enabling groups to cover spaces far beyond the capacity of any individual.

323 *The limits of pure exploitation.*

324 The biological picture motivates a broader claim for artificial agents: in open and
325 non-stationary worlds, pure exploitation is structurally self-limiting. An agent that
326 only selects actions that appear best under its current belief may obtain short-term
327 reward, but gradually loses the conditions required for continued adaptation. Once
328 it stops seeking information, it can no longer detect environmental change, discover
329 better strategies, or preserve access to regions from which future learning would have
330 come. This gives rise to three failure modes: **belief stagnation**, where the agent’s
331 model becomes stale under distributional shift; **value stagnation**, where acquired
332 information fails to translate into improved behaviour; and **reachability collapse**,
333 where the policy contracts onto familiar states and actions, foreclosing future learning
334 opportunities.

335 These failures appear across the AGI trajectory. In reasoning systems, they man-
336 ifest as unresolved uncertainty, reward plateaus, and entropy collapse; in embodied
337 agents, as stale representations under partial observability, unsafe exploration-
338 exploitation cascades, and limited skill expansion; in world models, as confusion
339 between epistemic uncertainty and irreducible noise, compounding imagination drift,
340 and loss of safety-critical detail; and in multi-agent systems, as rigid coordination
341 patterns that suppress alternative roles or communication structures.

342
343 *Classical RL exploration as a partial remedy.*

344 Reinforcement learning has long studied mechanisms for avoiding premature com-
345 mitment to currently high-value actions. However, classical exploration methods only
346 partially address the three limits above for agentic systems. Undirected stochastic
347 strategies such as ϵ -greedy and Boltzmann exploration inject randomness, but do not
348 track what the agent knows or whether exploration improves future value. Optimism-
349 based methods such as UCB prefer uncertain actions and therefore partially support
350 information acquisition and coverage, but usually rely on stationarity assumptions [9].
351 Posterior-sampling methods such as PSRL maintain a Bayesian belief, yet information
352 gain is still an indirect consequence rather than an explicit objective [10]. Prediction-
353 error bonuses such as ICM reward novelty, but do not guarantee policy improvement
354 and remain vulnerable to noisy-TV effects [11]. Ensemble-based approaches such as
355 EE-Net better approximate epistemic uncertainty through disagreement, but remain
356 largely confined to stationary single-task regimes [12]. Entropy-regularised methods
357 such as SAC help preserve policy stochasticity and resist premature collapse, but do
358 not determine where exploration should occur or why it should improve long-term
359 value.

360 Thus, classical RL provides important ingredients but leaves three structural gaps.
361 First, most methods assume stationarity, whereas open-world intelligence must handle
362 drift and changing objectives. Second, most operate over primitive actions, whereas
363 modern agents must explore over reasoning steps, tool calls, memory retrieval, world-
364 model rollouts, and coordination structures. Third, most lack metacognition: they do
365 not explicitly track what is unknown, what has become stale, or when renewed explo-
366 ration is needed. These gaps motivate a unified framework of epistemic exploration,

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in which an agent must jointly acquire information, convert information into durable improvement, and preserve the reachability required for future adaptation.

2.2 What Is Epistemic Exploration? — *A Unified Definition*

In this section, we define *epistemic exploration*. The term *epistemic* originates from epistemology—the study of knowledge and belief. We adopt this term because the essence of exploration is fundamentally an epistemic process: an agent explores in order to discover what it does not know, update its beliefs, and construct increasingly accurate models of the world. Exploration is therefore not merely a behavioural phenomenon, nor simply a manifestation of curiosity, but a mechanism for knowledge acquisition and refinement. From this perspective, we formalise epistemic exploration through three criteria and unify them into a single optimisation objective. An epistemic explorer actively acquires information that reduces uncertainty about the world, seek experiences at the boundary of its current capabilities and transform them into improved capabilities and policies, and preserves the ability to continue acquiring knowledge in the future.

2.2.1 Three-Criterion Definition

We briefly introduce the main notation used throughout this section. Most of it is standard and is collected here primarily for self-containedness.

Environment.

The environment is a Markov Decision Process $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, r, \gamma)$, with state space $\mathcal{S}$, action space $\mathcal{A}$, transition kernel $P : \mathcal{S} \times \mathcal{A} \rightarrow \Delta(\mathcal{S})$, reward distribution $r : \mathcal{S} \times \mathcal{A} \rightarrow \Delta(\mathbb{R})$, and discount factor $\gamma \in [0, 1)$. For each $\theta \in \Theta$, we write P_θ and r_θ for the transition kernel and reward distribution of the MDP parameterised by θ .

Definition 1 (Agent) An **agent** operating in environment $\mathcal{M}$ is a tuple $\mathfrak{A} = (\mathcal{H}, h_0, \mu, \pi_{\mathfrak{A}})$, where $\mathcal{H}$ is an internal state space encoding beliefs, knowledge, and memory; $h_0 \in \mathcal{H}$ is the initial internal state; $\mu : \mathcal{H} \times \mathcal{S} \times \mathcal{A} \times \mathbb{R} \times \mathcal{S} \rightarrow \mathcal{H}$ is the internal-state update rule ($h_{t+1} = \mu(h_t, s_t, a_t, r_t, s_{t+1})$); and $\pi_{\mathfrak{A}} : \mathcal{H} \times \mathcal{S} \rightarrow \Delta(\mathcal{A})$ is the agent policy ($a_t \sim \pi_{\mathfrak{A}}(\cdot \mid h_t, s_t)$). $\square$

The distinction between a policy and an agent is not merely notational. A stateless policy $\pi(a \mid s)$ is the special case $\mathcal{H} = \{\emptyset\}$ with trivial μ . The full agent formalism is needed because epistemic exploration is fundamentally about what happens *inside* the agent: how it accumulates information across time (memory, via h_t), how it refines its understanding from experience (belief tracking, via μ), and how it conditions action selection on both the environment state and its own epistemic status (goal-directed reasoning, via $\pi_{\mathfrak{A}}$).

Belief and epistemic uncertainty.

Although $\mathcal{M}$ is well-defined in principle, the agent may lack direct access to P and r . In the unknown-environment setting, the agent maintains a posterior belief $b_t(\theta) = p(\theta \mid$

$\mathcal{D}_t) \in \Delta(\Theta)$ over a parameter space Θ indexing plausible environments, where $\theta^* \in \Theta$ is the true parameter, b_0 is the prior, and $\mathcal{D}_t = \{(s_i, a_i, r_i, s_{i+1})\}_{i=0}^{t-1}$ is the trajectory data collected so far. The agent’s **information state** at time t is $\mathcal{I}_t = (h_t, s_t)$, jointly capturing internal memory and the current environment state.

The central quantity for epistemic exploration is the epistemic uncertainty at a state–action pair:

Definition 2 (Epistemic Uncertainty) The **epistemic uncertainty** at (s, a) under belief b_t is the mutual information between the environment parameter and the next transition:

$$\begin{aligned} \mathcal{U}(s, a; b_t) &= I(\theta; s_{t+1}, r_t \mid s_t = s, a_t = a, b_t) \\ &= H[p(s_{t+1}, r_t \mid s, a, b_t)] - \mathbb{E}_{\theta \sim b_t} [H[p(s_{t+1}, r_t \mid s, a, \theta)]] \end{aligned} \quad (1)$$

□

The first term measures total uncertainty, while the second measures irreducible randomness. Their difference captures *epistemic uncertainty*—the uncertainty that can be reduced through additional experience. Maximising $\mathcal{U}$ therefore encourages exploration of informative, rather than merely noisy, state–action pairs. Equivalently, $\mathcal{U}$ can be written as the expected Bayesian update to the belief:

$$\mathcal{U}(s, a; b_t) = \mathbb{E}_{(s_{t+1}, r_t) \sim p(\cdot \mid s, a, b_t)} [D_{\text{KL}}(b_{t+1}(\theta) \parallel b_t(\theta))], \quad (2)$$

where $b_{t+1}(\theta) = p(\theta \mid \mathcal{D}_t \cup \{(s, a, r_t, s_{t+1})\})$. In this form, $\mathcal{U}(s, a; b_t)$ is zero iff the outcome leaves the posterior unchanged in expectation, and strictly positive whenever different $\theta \in \text{supp}(b_t)$ predict different transition distributions at (s, a) .

For any belief b over plausible environments, the *Bayes value function* of policy $\pi_{\mathfrak{A}}$ is defined as

$$V_b^{\pi_{\mathfrak{A}}}(\mathcal{I}) := \mathbb{E}_{\theta \sim b} [V_{\theta}^{\pi_{\mathfrak{A}}}(s, h)], \quad \mathcal{I} = (h, s).$$

When the belief is the current posterior b_t , we write $V_{\text{Bayes}}^{\pi_{\mathfrak{A}}}(\mathcal{I}_t) \equiv V_{b_t}^{\pi_{\mathfrak{A}}}(\mathcal{I}_t)$. This value function reflects what the agent expects to achieve under its current belief, rather than what it would achieve if the true environment parameter θ^* were already known. The gap between $V_{b_t}^{\pi_{\mathfrak{A}}}(\mathcal{I}_t)$ and $V_{\theta^*}^{\pi_{\mathfrak{A}}}(\mathcal{I}_t)$ measures the cost of ignorance; reducing this gap through information acquisition and policy improvement is the purpose of epistemic exploration. We now define epistemic exploration through three jointly necessary criteria.

Definition 3 (Epistemic Exploration) Let $\mathfrak{A}$ be an agent operating in an environment parameterised by an unknown $\theta \in \Theta$, with posterior belief $b_t(\theta) = p(\theta \mid \mathcal{D}_t)$. The agent $\mathfrak{A}$ engages in *epistemic exploration* if $\mathfrak{A}$ simultaneously satisfies the following three criteria for some positive thresholds $\varepsilon_{t,1}, \varepsilon_{t,2}, \varepsilon_{t,3} > 0$.

Criterion 1: Information Gain. The agent’s exploratory actions are directed toward reducing epistemic uncertainty, i.e., actions target transitions with non-trivial expected belief

update. Formally, the policy selects actions with high epistemic uncertainty (Definition 2):

$$a_t \in \{a \in \mathcal{A} \mid \mathcal{U}(s_t, a; b_t) \geq \varepsilon_{t,1}\}.$$

Equivalently, by Eq. (2), the expected Bayesian update $\mathbb{E}[D_{\text{KL}}(b_{t+1}||b_t)]$ is at least $\varepsilon_{t,1}$. Exploration is directed **where the agent knows least**.

Running example. Consider a Mars rover searching for biosignatures. Its orbital map provides only coarse and partially outdated information, so the rover must decide which rock formation to inspect first. Criterion 1 directs it toward the site about which its current model is most uncertain—for example, an outcrop whose measurements are equally consistent with a mundane volcanic origin and a potentially water-related clay deposit. Inspecting that site is most informative because it would produce the largest belief update. Exploration is therefore directed not toward what already looks best understood, but toward what is most uncertain and most informative.

Criterion 2: Value Improvement. Exploratory actions must seek experiences at the boundary of the agent’s current capabilities and convert the resulting information into durable policy improvement. After the experience induced by the exploratory action updates the policy from $\pi_{\mathfrak{A},t}$ to $\pi_{\mathfrak{A},t+1}$, the updated policy should improve expected value relative to the previous policy:

$$\mathbb{E}\left[V_{b_{t+1}}^{\pi_{\mathfrak{A},t+1}}(\mathcal{I}_t) - V_{b_{t+1}}^{\pi_{\mathfrak{A},t}}(\mathcal{I}_t) \mid \mathcal{I}_t\right] \geq \varepsilon_{t,2}.$$

Here, both policies are evaluated from the same reference information state $\mathcal{I}_t$ under the same updated belief b_{t+1} . Evaluating the old policy under b_{t+1} provides a counterfactual baseline: it asks whether the newly acquired information has produced a better policy, rather than merely changed the agent’s belief or moved the agent to a different state. Exploration is directed **at what the agent cannot yet do well**.

Running example. Consider a household robot learning to prepare meals in a new kitchen. Its belief state encodes which manipulation skills it has already mastered: opening drawers, picking up rigid objects, pouring water, or placing items on a counter. Criterion 2 governs which tasks the robot should attempt next. Repeatedly picking up cups or opening the same drawer lies well inside its current competence and produces little improvement. Attempting to fillet fish or operate an unfamiliar gas stove may lie far beyond its competence frontier, producing unsafe failures or demonstrations that cannot yet be converted into a generalisable skill. The most valuable exploratory tasks are those at the frontier of current ability: for example, opening a slightly heavier cabinet, pouring liquid into a narrower container, or using a familiar gripper to handle a mildly deformable object. These tasks are challenging enough to update the robot’s beliefs about its own manipulation limits, yet tractable enough for the updated policy to convert the experience into durable capability improvement.

Criterion 3: Epistemic Reachability. The agent must preserve access to regions that remain feasible, belief-consistent, and epistemically relevant under its current information state. Let

$$d_{b_t}^{\pi}(s, a) = \mathbb{E}_{\theta \sim b_t} \left[(1 - \gamma) \sum_{k=0}^{\infty} \gamma^k \Pr_{\pi, P_{\theta}}(s_k = s, a_k = a) \right]$$

denote the belief-averaged discounted visitation of (s, a) under policy π . Let $\mathcal{R}_t \subseteq \mathcal{S} \times \mathcal{A}$ denote the set of state–action regions that are currently feasible, safe, belief-consistent, and potentially useful for future information gain or capability improvement. Then $\pi_{\mathfrak{A},t}$ should satisfy

$$d_{b_t}^{\pi_{\mathfrak{A},t}}(s, a) \geq \varepsilon_{t,3}(s, a), \quad \forall (s, a) \in \mathcal{R}_t,$$

where $\varepsilon_t(s, a) \geq 0$ is a possibly state–action-dependent reachability threshold. Exploration is directed **where the agent might otherwise prematurely lose access**.

Running example. Consider an AI research agent exploring a large codebase to diagnose a system failure. After finding one suspicious module, it begins to focus almost exclusively on that part of the repository, repeatedly reading nearby files, rerunning similar tests, and refining the same hypothesis. Meanwhile, it stops visiting several other parts of the codebase—for example, an old configuration pipeline and a rarely used data-preprocessing script—because they do not seem immediately relevant. Suppose, however, that one of these neglected components is exactly where the hidden bug originates, and that later changes to logs, cached states, or temporary files will make that evidence much harder to recover. If the agent drops those paths too early, it permanently loses the opportunity to discover the true cause. Criterion 3 therefore requires the agent to preserve sufficient coverage over plausible but currently underexplored regions of the environment, so that future learning remains possible rather than being foreclosed by premature commitment. $\square$

2.2.2 Unified Epistemic Exploration Objective

The three criteria can be unified into a single optimisation problem:

$$\pi_{\mathfrak{A},t}^* = \underbrace{\arg \max_{\pi_{\mathfrak{A}} \in \Pi_{\text{reach}}(b_t, \mathcal{R}_t)}}_{\text{Epistemic Reachability (C3)}} \underbrace{\mathbb{E}_{\pi_{\mathfrak{A}}, b_t} [V_{b_{t+1}}^{\pi_{\mathfrak{A},t+1}}(\mathcal{I}_t) - V_{b_{t+1}}^{\pi_{\mathfrak{A},t}}(\mathcal{I}_t)]}_{\text{Value Improvement (C2)}} + \underbrace{\beta \cdot \mathbb{E}_{\pi_{\mathfrak{A}}, b_t} \left[\sum_{t'=t}^{\infty} \gamma^{t'-t} \mathcal{U}(s_{t'}, a_{t'}; b_{t'}) \right]}_{\text{Information Gain (C1)}}. \quad (3)$$

The second term in Eq. (3) operationalises Criterion 1 by rewarding trajectories that are expected to reduce epistemic uncertainty. The expectation $\mathbb{E}_{\pi_{\mathfrak{A}}, b_t}[\cdot]$ is taken over trajectories generated by executing $\pi_{\mathfrak{A}}$ from $\mathcal{I}_t$, with environment dynamics averaged under the current belief b_t . The scalar $\beta > 0$ controls the relative strength of information-seeking compared with value improvement. The first term in Eq. (3) measures the expected counterfactual Bayes-value improvement induced by the experience collected under $\pi_{\mathfrak{A}}$ (Criterion 2). The feasible policy set encodes Criterion 3:

$$\Pi_{\text{reach}}(b_t, \mathcal{R}_t) := \left\{ \pi_{\mathfrak{A}} \mid d_{b_t}^{\pi_{\mathfrak{A}}}(s, a) \geq \varepsilon_{t,3}(s, a), \quad \forall (s, a) \in \mathcal{R}_t \right\}.$$

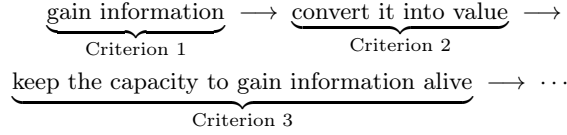
Thus, epistemic reachability does not require uniform coverage of all reachable state–action pairs; rather, it prevents the agent from prematurely losing access to regions that may still matter for future learning or adaptation.

Why three criteria and not fewer?

The information-gain term asks whether the same experience will reduce uncertainty about the world. The value-improvement term asks whether the experience induced by a policy will move the agent beyond its current capability boundary and lead to a better future policy. The reachability constraint asks whether the policy preserves access to plausible future states, actions, and strategies. In this sense, epistemic exploration

is not random novelty seeking; it is structured information acquisition that improves capability while keeping future adaptation possible. An agentic system satisfying only Criterion 1 might spend its exploration budget on regions of the state space that are informationally rich but strategically worthless; a system satisfying only Criterion 2 might improve locally while its global model degrades unnoticed; a system satisfying only Criterion 3 might maintain broad coverage without ever extracting useful information from it.

Remark 1 (Joint necessity) The three criteria form a single closed loop, not an arbitrary list:



The loop is self-sustaining when all three hold and self-destructing when any one is removed. In particular, Criterion 3 is the enabling condition for Criterion 1: an agent that forecloses on belief-consistent regions cannot generate the transitions that would update its beliefs there. $\square$

2.3 How to Explore: *Signal-Driven Methodologies*

We now briefly summarise *how* these requirements are operationalised in modern AI systems. While the unified objective in Eq. 3 provides a normative target, practical systems rely on computable signals that decide when an agent should depart from pure exploitation. We organise these signals into three methodological families, each corresponding to one criterion of epistemic exploration.

1. Uncertainty-driven exploration. This methodology operationalises Information Gain. The agent explores where its predictions are least reliable, using signals such as entropy, ensemble disagreement, predictive variance, confidence margins, or posterior uncertainty. In reasoning models, this appears as entropy-guided branching over uncertain tokens or reasoning steps; in embodied agents, as active perception and uncertainty-aware navigation; in world models, as disagreement-driven probing of poorly modelled latent regions; and in multi-agent systems, as exploration driven by inter-agent disagreement or joint-belief uncertainty.

2. Competence-driven exploration. This methodology operationalises Value Improvement. The agent explores tasks or states near the boundary of its current capability, where learning progress is most likely. Instead of sampling arbitrary novelty, it prioritises experiences that are neither already mastered nor hopelessly difficult. In practice, this includes curriculum learning, difficulty-aware sampling, self-play, failure-conditioned data generation, skill bootstrapping, and imagination-based skill discovery. Such mechanisms ensure that acquired information is converted into durable improvements in reasoning, acting, planning, or coordination.

3. Reachability-driven exploration. This methodology operationalises Epistemic Reachability. The agent preserves access to diverse states, actions, strategies, reasoning paths, tools, latent futures, or coordination structures before they are

prematurely excluded by exploitation. In practice, this includes entropy regularisation, coverage bonuses, beam or trajectory diversity, anti-collapse constraints, Go-Explore-style return-to-promising-state mechanisms, latent-space diversity bonuses, and role-diversity incentives in multi-agent systems. Reachability-driven exploration is especially important because it maintains the conditions under which future information gain and value improvement remain possible.

These three methodologies are complementary rather than mutually exclusive. Uncertainty-driven exploration identifies *where agents know least*; competence-driven exploration identifies *where agents can't yet do well*; and reachability-driven exploration identifies *where agents might never go*.

3 Exploration Across the Five Levels Toward AGI

Chapter 2 established the formal foundations of epistemic exploration. This chapter asks: *how does this framework instantiate across the trajectory from narrow AI to AGI?* We organise the answer around five levels and their corresponding exploration spaces. Level 1 serves as the reactive baseline, while Levels 2–5 are defined by the new exploration capacity required to move beyond the previous level:

1. **Level 2: Reasoner — reasoning-space exploration.** A responder mainly relies on learned input–output mappings and local token-level variation, with minimal explicit epistemic exploration. The transition to a reasoner opens reasoning space as an object of exploration: token sequences, reasoning trajectories, and latent representations become searchable structures. The system explores for informative reasoning paths rather than merely producing a single reactive response.
2. **Level 3: Agent — interaction-space exploration.** Moving from reasoner to agent requires exploration beyond internal deliberation. Interaction-space exploration extends intelligence into perception and action: planning, tool invocation, memory management, active perception, and closed-loop interaction in digital or physical environments.
3. **Level 4: Prospector — imagination-space exploration.** Moving from agent to prospector requires exploration beyond real interaction. Imagination-space exploration shifts the search into learned world models, where the agent constructs, interrogates, and refines imagined futures before committing to costly or risky real-world actions.
4. **Level 5: Ecosystem — coordination-space exploration.** Moving from prospector to ecosystem requires exploration beyond the individual agent. Coordination-space exploration makes communication protocols, collaboration topologies, role assignments, shared representations, and collective knowledge substrates the objects of exploration.

3.1 Level 1–2: Responder → Reasoner — reasoning-space exploration

As **responders** move beyond scripted Q&A and template-based dialogue, the transition toward **reasoners** is driven by the emergence of exploration within the reasoning process. Traditional responders map inputs to outputs in a single forward pass, producing fluent responses but lacking internal search, verification, revision, or backtracking. They largely interpolate within the training distribution, whereas real-world tasks require identifying missing information, testing uncertain assumptions, and updating plans as evidence changes. A **reasoner**, by contrast, explores reasoning space: it branches over hypotheses, evaluates intermediate steps, revises failed paths, and selects trajectories that reduce uncertainty and improve correctness.

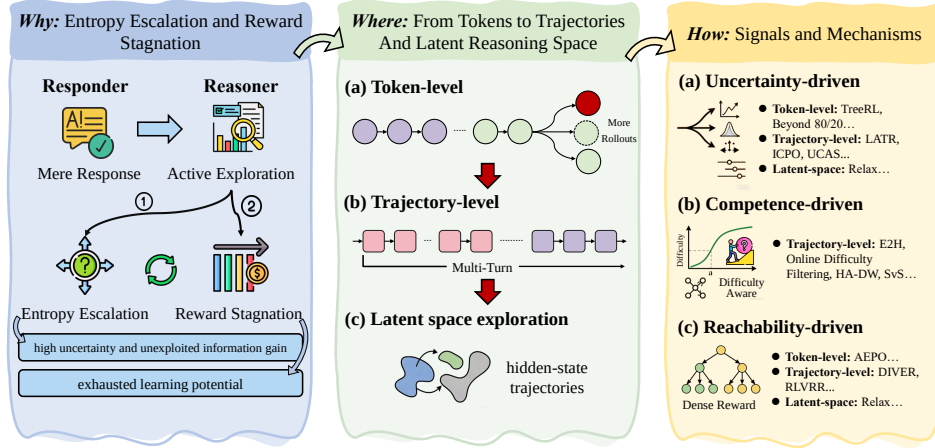


Fig. 3: Why–Where–How overview of Level 2 exploration in reasoners. **Why:** entropy escalation and reward stagnation motivate exploration for resolving uncertain reasoning decisions, escaping narrow trajectory distributions, and sustaining further improvement. **Where:** token-level decisions, complete reasoning trajectories, and latent reasoning states define the reasoning space in which exploration is instantiated. **How:** uncertainty-driven, competence-driven, and reachability-driven mechanisms operate across these reasoning locations.

3.1.1 Why: Entropy Escalation and Reward Stagnation

The shift from responders to reasoners marks a move from passive generation to active reasoning-space exploration. While responders mainly rely on pretrained patterns, reasoners must search for missing information, test hypotheses, and refine decisions under uncertainty. We therefore analyze exploration in reasoning systems through two main signals: entropy, which reflects uncertainty and exploration breadth, and reward, which guides exploration direction and effectiveness.

Entropy Escalation. Entropy is a key signal of uncertainty in reasoning-space exploration: higher policy entropy indicates that the model has not formed stable preferences among competing reasoning trajectories. Prematurely exploiting one path may therefore cause persistent errors or suboptimal convergence. Exploration is needed to broaden the search, reduce uncertainty, and identify more reliable decision patterns. This aligns with active learning [13], where high-entropy regions often correspond to higher information gain.

Reward Stagnation. Rewards guide the direction of exploration, but stagnant reward improvement often indicates that the policy is trapped in a familiar behavioral mode rather than mastering the task. The model may keep interpolating within known regions while failing to discover trajectories that provide new feedback [14]. Thus, reward stagnation signals insufficient exploration and motivates expanding the trajectory distribution.

Together, entropy escalation and reward stagnation provide complementary signals: entropy indicates where uncertainty remains unresolved, while reward stagnation indicates when current reasoning trajectories have become too narrow to yield further improvement.

3.1.2 Where: From Tokens to Trajectories and Latent Reasoning Spaces

Where exploration occurs in the reasoner depends on where actions are defined. This choice is not merely a matter of terminology: it determines what the policy can control, what feedback can evaluate, and how expensive search becomes. In conventional language generation, actions are token-level decoding decisions, enabling local exploration through next-token choices and entropy-guided branching. At a coarser granularity, actions can be treated as complete sampled reasoning trajectories, allowing the system to compare, diversify, and optimize full solution paths under verifiable rewards. More recently, exploration has moved beyond surface-level discrete actions into latent reasoning spaces, where the agent explores hidden-state trajectories or latent actions before committing to natural-language tokens. These three locations define increasingly abstract levels of reasoning-space exploration: local branching choices, complete reasoning paths, and latent reasoning states.

Token Level. At the token level, the explored object is the local decision point inside the next-token distribution. This granularity is useful when uncertainty is concentrated at a few decisive positions, because a single token may change the direction of the following proof, program, or explanation. Beyond the 80/20 Rule identifies high-entropy tokens as forking tokens that determine the direction of subsequent reasoning [2]. TreeRL further turns this view into a concrete search procedure by traversing the current reasoning tree, selecting the Top- N high-entropy tokens or steps, and branching from these positions [15]. Token-level exploration therefore focuses on how local uncertain decisions reshape the downstream reasoning path, while leaving open whether local variation actually produces globally distinct solutions.

Trajectory Level. At the trajectory level, the explored object is a complete chain-of-thought or solution rollout, which is also the unit that receives final rewards and

determines task success. This makes trajectories the natural granularity for verifiable-reward training, where the system must compare complete solution paths rather than isolated decoding choices. Local token stochasticity does not necessarily produce genuinely different reasoning trajectories. LATR explicitly identifies this limitation: token-level sampling can collapse into near-identical paths, motivating lookahead tree-based rollouts that enforce trajectory-level diversity [16]. DIVER reaches a similar conclusion from the perspective of global sequence-level diversity, arguing that effective exploration should cover semantically different solution paths rather than merely perturbing local tokens [17].

Latent-Space Level. Reasoning-space exploration need not be fully externalized as natural-language tokens. At this level, exploration occurs before discretization, allowing the model to search over hidden states or compact latent actions that may summarize several possible continuations. Coconut feeds the model’s hidden state back as a continuous thought, allowing a latent state to encode multiple possible next reasoning steps before committing to a discrete token sequence [18]. CoLA introduces a compact latent action space so that RL can operate over more controllable and semantically structured actions rather than directly over token actions [19]. ReLaX further treats hidden-state dynamics as an explicit object for regulating exploration and exploitation, showing that latent reasoning space contains measurable exploratory structure beyond surface token entropy [4]. Compared with token or trajectory exploration, latent-space exploration is less directly interpretable, but it can expose exploratory structure before it appears in surface text.

Table 1: Level 2 Reasoning-Space Exploration Organized by Where and How

Where \ How	Uncertainty-driven	Competence-driven	Reachability-driven
Token-level	High-entropy tokens expose local branching points: Beyond 80/20 [2], TreeRL [15].	–	Entropy control preserves exploratory probability mass in token distributions: AEPO [20].
Trajectory-level	Confidence and rollout diversity guide exploration over complete reasoning paths: LATR [16], ICPO [21], UCAS [22].	Difficulty scheduling and self-synthesis keep exploration near the competence frontier: E2H [3], Online Difficulty Filtering [23], HA-DW [24], SvS [25].	Global diversity and smoother rewards preserve alternative reasoning paths: DIVER [17], RLVRR [26].
Latent-space level	Latent dynamics reveal exploratory structure beyond surface token entropy: ReLaX [4].	–	Latent exploration regulates premature determinism in hidden-state dynamics: ReLaX [4].

3.1.3 How: Signals and Mechanisms

Following the three-methodology framework of Section 2.3, we organize the reasoner’s exploration mechanisms along the three criterion-aligned axes: uncertainty-driven, competence-driven, and reachability-driven. In reasoning systems, entropy and reward

783 serve as two core control signals that cut across these methodologies: entropy reflects
 784 the uncertainty and diversity of current decisions, while reward, shaped by external
 785 objectives and verifiers, determines the direction and effectiveness of exploration.
 786 Together, they provide a scheduling principle that elevates exploration from an
 787 outcome measure to a controllable mechanism.

788 **Uncertainty-driven Exploration.** In language reasoning models, uncertainty-
 789 driven exploration mainly uses entropy and confidence signals to decide where
 790 reasoning should branch and how much stochasticity should be preserved. This is the
 791 Level 2 form of information gain: uncertainty is useful when it points to decisions
 792 that can reveal better reasoning paths. At the token level, high-entropy positions act
 793 as local reasoning forks. Beyond the 80/20 Rule identifies such tokens as disproportionately
 794 important for reasoning performance, while TreeRL traverses the current
 795 reasoning tree, selects high-uncertainty tokens or intermediate steps, and expands new
 796 branches from these positions [2, 15]. Here, uncertainty decides where the next local
 797 branch should be opened.

798 At the trajectory level, uncertainty must be evaluated over complete sampled
 799 solutions rather than isolated token choices. LATR encourages lookahead trajectory
 800 diversity, ICPO uses relative generation probabilities as an intrinsic confidence signal,
 801 and UCAS combines response-level and token-level uncertainty to amplify correct-
 802 but-uncertain trajectories [16, 21, 22]. These works also clarify why entropy alone
 803 is insufficient: it measures distributional diffuseness, but not whether exploration is
 804 structured, diverse, and aligned with long-horizon reasoning.

805 At the latent-space level, uncertainty can be measured before reasoning is fully
 806 verbalized. ReLaX uses Dynamic Spectral Dispersion to quantify the heterogeneity
 807 of hidden-state dynamics, revealing exploratory structure that surface token entropy
 808 may miss [4]. This moves uncertainty-driven exploration from visible branches to the
 809 latent trajectories that precede them.

810 **Competence-driven Exploration.** For reasoning tasks, a model’s competence
 811 can be naturally measured by its final reward or success rate. As training progresses,
 812 competence evolves and gradually unlocks new reward regions for exploration. This
 813 motivates competence-driven exploration, where training signals and data selection
 814 are adapted to the learner’s current ability, focusing search where improvement is still
 815 attainable. We do not isolate token-level or latent-space competence as separate categories,
 816 because competence is usually observed through trajectory-level success, final
 817 reward, or sample difficulty rather than a single token or hidden state. At the trajectory
 818 level, a common strategy is to keep sampled problems near the current capability
 819 frontier. E2H decomposes training data into increasing difficulty stages and schedules
 820 them from easy to hard, allowing simpler tasks to build reusable skills before harder
 821 tasks dominate training [3]. Online Difficulty Filtering selects medium-difficulty samples
 822 according to current success rates, while HA-DW corrects the difficulty-dependent
 823 bias of group-relative advantages [23, 24]. SvS uses correct solutions on underperforming
 824 samples to synthesize answer-preserving variants, expanding the training frontier
 825 and sustaining generation diversity [25].

826 **Reachability-driven Exploration.** Aligned with Criterion 3 (Epistemic Reachability),
 827 reachability-driven exploration focuses on preventing the reasoning policy
 828

from collapsing too early onto a narrow set of trajectories. At the token level, this means preserving local branches that might otherwise disappear after aggressive policy updates. AEPO regulates policy entropy around a target level, keeping exploratory probability mass in the token distribution while avoiding unconstrained randomness [20]. This makes uncertain token branches revisitable in later optimization.

At the trajectory level, reachability depends on keeping alternative solution paths viable. DIVER introduces global sequence-level diversity incentives so that the policy continues to visit semantically distinct rollouts, while RLVR uses reward chains to smooth supervision and reduce the sharp reward boundaries that can prematurely eliminate partially useful trajectories [17, 26]. In this view, reward design is part of the infrastructure that keeps the reasoning space open.

At the latent-space level, reachability concerns whether hidden-state trajectories remain diverse before they collapse into deterministic surface forms. ReLaX uses DSD-based regularization to counteract rigid internal computation and premature convergence, motivating reachability preservation beyond visible token distributions [4]. The reachable space of a reasoner therefore includes not only sampled answers, but also the latent continuations from which those answers emerge.

3.2 Level 3: Agent — Interaction-Space Exploration

At Level 3, the agent crosses a qualitative boundary: it moves from purely internal reasoning into situated interaction with external environments. The “Reasoner” of Level 2 generates hypotheses within its own token space; the “Agent” must act on those hypotheses—querying databases, invoking APIs, reorienting sensors, or manipulating physical objects—and update its beliefs from the consequences. Exploration can no longer be confined to branching over reasoning trajectories; it now unfolds in perception and action space, where every exploratory step incurs real cost (latency, API calls, energy, or potentially irreversible physical consequences).

This level encompasses two qualitatively distinct domains that share the same formal framework but differ in action space, feedback structure, and risk profile:

- **Digital agents** operate in software-mediated environments—web browsers, code interpreters, file systems, and tool APIs, as shown in Figure 4. Their action space is discrete and combinatorial (which tool to call, with what arguments, in what order), feedback is fast but often sparse, and the primary exploration challenge is discovering effective tool-invocation policies across a vast combinatorial manifold.
- **Embodied agents** operate in physical or high-fidelity simulated environments via robotic effectors and multimodal sensors, as shown in Figure 5. Their action space is continuous and high-dimensional, feedback is slow and expensive, and the primary exploration challenge is safely building grounded scene knowledge while expanding embodied competence under severe partial observability, costly feedback, and potentially irreversible physical consequences.

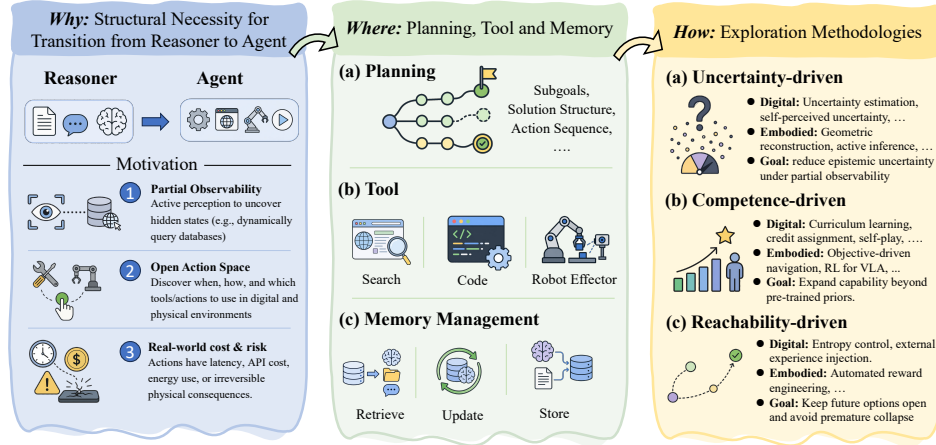


Fig. 4: A Why, Where, How overview of agentic exploration at Level 3, where the system transitions from a passive reasoner to an agent that can actively interact with the external world.

3.2.1 Why: The Inevitability of Exploration in the Transition to Autonomous Agency

The transition toward Level 3 AGI requires a shift from passive reasoning to autonomous, situated agency. Modern reasoners can solve conversational tasks, but they mostly react within fixed interfaces and pre-trained priors. Autonomous agents must instead coordinate planning, tool use, memory, perception, and action while operating in open-ended digital and physical environments [27–30]. In such settings, exploration is not an auxiliary optimization; it is the mechanism that lets an agent uncover hidden state, discover executable strategies, and expand competence beyond static pre-training.

Resolving Partial Observability through Active Evidence Acquisition. Both digital and embodied agents face information that is hidden behind interfaces, long interaction histories, unseen geometry, or latent physical properties. Passive observation is therefore insufficient: the agent must actively query, retrieve, move, inspect, or manipulate to convert uncertainty into usable evidence. Digital agents reduce uncertainty through memory retrieval, posterior sampling, and knowledge-seeking behaviours [29, 31, 32]; embodied agents reduce it by selecting informative views, reconstructing scenes, storing frontier memories, or physically testing object properties [33–35]. Exploration thus becomes the bridge between internal reasoning and externally grounded state estimation.

Mastering Combinatorial Action Spaces through Structured Interaction. Agency also expands the action space. A digital agent must decide when to call tools, which APIs or functions to use, how to parameterize them, and how to integrate observations into multi-step plans. An embodied agent faces analogous choices over frontiers, subgoals, routes, affordances, and motor behaviours. Exhaustive search is impossible, so the agent must explore structured decision spaces rather than raw

action manifolds. Tool-use and planning methods explore function-calling paths, off-policy traces, curricula, and self-verified reasoning trajectories [28, 36–38]; embodied methods explore semantic frontiers, language-grounded search regions, reusable skills, and plan repairs before execution [39–41].

Expanding Competence under Costly and Irreversible Interaction. Unlike text-only reasoning, actions in software systems and physical environments can consume resources, alter state, or cause irreversible failures. Naive stochastic exploration is therefore inadequate. The agent needs competence-driven and reachability-preserving exploration: curricula to approach harder tasks, credit assignment to identify useful intermediate decisions, self-verification to repair trajectories, and diverse behaviour repertoires to keep future options open [36, 42, 43]. In embodied settings, this includes learning from corrections, fine-tuning action policies through interaction, sharing experience across tasks, and using tasks as resets for one another [44–47]. Exploration is therefore the structural condition for sustained autonomous agency.

3.2.2 Where: Planning, Tool Usage, and Memory Management in Digital Agents

For digital agents, exploration is instantiated through the computational interfaces that mediate interaction with digital environments. We organize these interfaces into three functional locations: *planning*, *tool usage*, and *memory management*.

Trajectory Planning. Planning provides the decision structure through which a digital agent decomposes tasks, selects intermediate actions, revises strategies, and coordinates multi-step interaction with an environment. Exploration at this level concerns how the agent searches over possible reasoning paths, subgoals, task curricula, and trajectory-level policies before or during execution [27, 28, 38, 43, 48, 49]. For example, PilotRL progressively expands the planning problem from plan following to plan generation and planner–executor coordination [28]. KnowSelf instead explores at the level of planning-mode selection, routing the agent among direct execution, reflection, and knowledge invocation according to self-perceived uncertainty [27].

Tool Interaction. Tool usage provides the executable interface through which a digital agent extends its native language-model capability with external computation and environment interaction. This includes deciding when to call a tool, which tool to invoke, how to parameterize the call, and how to integrate returned observations into the ongoing reasoning trajectory. Exploration at this level concerns the discovery of effective tool-use behaviours, including diverse function-calling paths, retrieved off-policy traces, expert-guided tool-use branches, and self-verified tool-integrated reasoning [36, 37, 50, 51]. For example, Agent0-VL alternates Solver and Verifier roles so that tool calls are not only used for solving visual reasoning problems, but also for generating critiques and repair signals [36]. RAPO expands tool-use exploration by retrieving off-policy step-level traces during rollout, allowing the agent to condition on behaviours beyond its own on-policy samples [37].

Memory Management. Memory management provides temporal continuity for digital agents by preserving information beyond the immediate context window. It determines what interaction history, retrieved evidence, posterior belief, or compressed

summary should be stored and how this information should condition future decisions. Exploration at this level concerns how agents use memory-like structures to reduce uncertainty, extend long-horizon reasoning, and reuse prior experience across episodes or trajectory segments [29, 31, 32, 42]. For example, JitRL maintains a non-parametric repository of past state–action–return experiences and retrieves relevant memories to estimate just-in-time action advantages [29]. PSRL Agent maintains a textual posterior over possible environment dynamics and rewards, then samples from this posterior to guide exploration in the next episode [31].

Table 2: Where–How Mapping of Level 3: Digital Agent Exploration Methods

Where / How	Uncertainty-driven	Competence-driven	Reachability-driven
Tool usage	–	Use tool-grounded self-play and self-verification to improve tool-integrated reasoning: Agent0-VL [36].	Entropy regularization, retrieved off-policy traces, or expert-prefix branching: EGPO [50], RAPO [37], E ³ -TIR [51].
Planning	Route the agent’s next planning mode according to self-perceived uncertainty: KnowSelf [27].	Curriculum learning, fine-grained credit assignment, or self-generated task curricula: PilotRL [28], EAPO [48], GiGPO [38], Absolute Zero [49].	Regulate trajectory-level entropy during multi-turn planning: EPO [43].
Memory management	Use memory-like evidence to augment action probabilities, sampling textual posteriors, or retrieving and refining external knowledge: JitRL [29], PSRL Agent [31], Search-o1 [32].	Learn to compress long interaction histories into reusable summaries for continued exploration: ReSum-GRPO [42].	–

3.2.3 How: Exploration in Digital Agents

Digital agents—web agents, coding assistants, tool-augmented reasoners—explore in a discrete, combinatorial action space defined by tool invocations, API calls, and multi-step software interactions. The three methodologies take characteristic forms in this domain.

Interaction-driven exploration in digital agents is typically formalized as reinforcement learning or bandit-style optimization, where policies are iteratively refined through environment feedback under limited interaction budgets, balancing exploration efficiency and reward acquisition [52–54].

Uncertainty-Driven. Under partial observability, a digital agent must actively acquire information to resolve the epistemic uncertainty that arises when static pre-trained priors prove insufficient in open-ended environments. Uncertainty-driven exploration leverages uncertainty signals to prioritize informative interactions and mitigate overconfident failures. Existing work can be organized into explicit methods [29, 31] and implicit methods [31, 32]. *Explicit methods* typically provide a formal estimation of state uncertainty, using these metrics to directly augment action

probabilities. For example, JitRL [29] maintains a dynamically evolving repository of historical experiences to estimate advantages and adjust action probability on-the-fly. [31] propose an LLM-based PSRL agent, where exploration is induced by maintaining a textual posterior over the unknown environment, sampling a plausible environment hypothesis before each episode, and acting optimally under this sampled hypothesis. *Implicit methods*, in contrast, internalize the uncertainty-to-augmentation pipeline into a self-awareness schema, enabling the agent to estimate its own capability boundaries and adaptively trigger strategic behaviors, such as reflection or external knowledge seeking. For example, KnowSelf [27] lets the agent assess whether a state exceeds its capability boundary to branch its execution among direct answering, internal reflection, or external knowledge invocation.

Competence-Driven. To tame the combinatorial action spaces inherent in agentic interaction, structured strategic exploration is essential. Systems must transcend naive trial-and-error, where exhaustive exploration in the raw action manifold is computationally intractable. Consequently, modern research in agentic RL focuses on strategic exploration—structuring the learning process to prune the action manifold and prioritize high-utility trajectories. We categorize these strategic paradigms into: 1) curriculum learning, which restricts the searchable action space through staged complexity; 2) prior-based credit assignment, which guides exploration via objective-aligned process signals; and 3) self-play, which autonomously generates a targeted distribution within the action manifold.

Curriculum learning methods typically adopt an easy-to-hard training pipeline, where the agent is first guided by explicit instructions or simplified sub-tasks before progressively handling complex, autonomous reasoning scenarios. For example, PilotRL [28] decomposes agentic decision-making into a global planner and a grounded executor, and adopts a three-stage progressive reinforcement learning curriculum to stabilize long-horizon training. The curriculum gradually shifts the learning focus from plan-following to plan-generation and finally to end-to-end coordination. Similarly, ReSum-GRPO [42] adopts a simple curriculum for long-horizon search.

Prior-based credit assignment methods reshape the optimization objective so that intermediate decisions receive objective-aligned learning signals, effectively steering exploration toward trajectories that are more likely to improve future performance. Concretely, they often augment sparse outcome rewards with structured intermediate terms that score process quality, such as the appropriateness and completeness of tool-invocation sequences or penalties for invalid or unnecessary tool calls. For example, EAPO [48] designs a fine-grained exploration reward to estimate the informational utility of each intermediate action. It treats exploration and memory as latent variables and uses variational inference to learn a proxy ($q_\phi(e, m \mid s)$) for the posterior distribution of successful exploration-memory configurations. GiGPO [38] improves agentic RL credit assignment by grouping identical (or matched) intermediate states across rollouts to estimate state-level relative advantages, enabling finer-grained updates beyond trajectory-level GRPO without extra environment sampling.

Self-play methods address the challenges of the vast action space by constructing a closed-loop mechanism that autonomously defines and refines the agent’s own visitation distribution [36, 49]. For example, Agent0-VL [36] achieves continual

improvement via a Self-Evolving Reasoning Cycle (SERC). A single unified policy alternates between two internal roles: a Solver that performs multi-turn tool-integrated reasoning and a Verifier that re-invokes tools to critique intermediate steps, producing fine-grained self-generated process rewards and confidence-gated repair signals. These self-generated rewards are then used to update the same shared policy with GRPO in an outer-loop optimization, enabling zero-external-reward self-improvement rather than curriculum-based staged training. More generally, exploration signals can be aggregated from heterogeneous or weak agents, where collaborative learning frameworks leverage diverse and imperfect feedback sources to enhance policy robustness and expand the effective exploration space [55].

Reachability-Driven. Maintaining behavioural flexibility is critical because a digital agent’s policy can rapidly collapse onto a narrow set of favoured tool-invocation patterns, foreclosing alternative interaction strategies before they can be evaluated. Reachability-driven methods explicitly optimise for sustained trajectory diversity to avoid premature convergence and to increase the chance of discovering higher-return interaction programs. Existing methods can be broadly grouped into two categories of entropy control methods and external experience injection methods.

Entropy control methods. This line of work aims to preserve behavioural flexibility by explicitly regulating policy entropy during training. For example, EGPO [50] incorporates an entropy bonus into advantage estimation over CoT tokens, encouraging high-entropy exploration over function-call trajectories during RL and mitigating early convergence to suboptimal tool-call patterns. EPO [43] identifies an exploration–exploitation cascade failure in multi-turn agent RL, where entropy cannot be independently controlled across turns because the policy parameters are shared. To address this issue, it combines trajectory-level entropy regularization, an entropy smoothing regularizer, and an adaptive phase-based weighting schedule to stabilize entropy dynamics throughout training, with stronger correction applied when the current entropy deviates excessively from historical levels.

External experience injection methods. This line of work seeks to enlarge the agent’s effective exploration space by exposing it to useful behaviours beyond its own on-policy rollouts, thereby mitigating the tendency to remain trapped within its native reasoning patterns [37, 51]. For example, RAPO [37] argues that pure on-policy Agentic RL restricts exploration to self-generated trajectories, and therefore introduces retrieval-augmented policy optimization, where the agent continuously reasons over retrieved off-policy step-level traces during rollout. By coupling a hybrid-policy rollout strategy with retrieval-aware policy optimization, including retrieval reward and importance shaping, RAPO explicitly broadens step-level exploration while preserving training stability.

3.2.4 Where: Perception, Planning, and Action in Embodied Agents

Embodied agents—mobile robots, robotic manipulators, and vision-language-action (VLA) controllers—explore by changing the external data-generating process. A camera pose, map frontier, subgoal, or motor command changes what evidence becomes observable, which states become reachable, and which behaviours can be improved.

For embodied systems, the “where” of exploration is therefore the carrier on which exploration is instantiated inside the perception–planning–action loop.

Perceptual Evidence. The carrier of perceptual exploration is sensor-grounded evidence: viewpoints, sensor poses, object observations, partial maps, scene memories, frontier snapshots, and physical inquiry results. Its form is active evidence acquisition. The agent selects camera poses or movement trajectories to expose unseen regions, reconstruct geometry, discover objects, test physical attributes, and store evidence that can be reused by later decisions. EVE defines this as embodied visual exploration over informative views and trajectories [30]. EXCALIBUR turns manipulation into perceptual inquiry because physical attributes such as mass and material require exploration and interaction rather than passive observation alone [35]. ActiveSplat explores by sampling and executing candidate views for online 3D reconstruction, while 3D-Mem represents explored regions as memory snapshots and unexplored regions as frontier snapshots [33, 34].

Embodied Planning. The carrier of planning exploration is an executable decision variable: a frontier, semantic region, route, subgoal, task, plan repair, affordance choice, or candidate action. Its form is pre-execution search, ranking, and revision. The agent does not merely write an internal plan; it chooses which embodied option should be tried next before paying the cost of movement or manipulation. SemExp, PONI, CoW, and VLFM explore over semantic frontiers, potential maps, open-vocabulary object-localization cues, and VLM-valued frontiers to decide where to search next [39, 56–58]. Voyager explores by choosing curriculum goals and reusable skills in an open world, DEPS explores by explaining failures and reordering candidate subgoals, and V-GPS explores at deployment time by reranking candidate robot actions before execution [40, 41, 59].

Behavioral Execution. The carrier of action exploration is the executed behaviour distribution: motor commands, action chunks, contact sequences, recovery trajectories, reset transitions, and task-conditioned skills. Its form is physical trial, correction, policy fine-tuning, and repertoire expansion under real or simulated dynamics. HIL-SERL interleaves autonomous rollouts with human corrections, so mistakes made during real-robot training become off-policy learning data [44]. DPPO explores structured action trajectories by fine-tuning diffusion policies with reinforcement learning [45]. MT-Opt shares exploration, experience, and representations across robot tasks, while Reset-Free Multi-Task RL uses different tasks as resets for one another [46, 47].

3.2.5 How: Exploration in Embodied Agents

After locating embodied exploration across perception, planning, and action, the three exploration methodologies can be read as orthogonal mechanisms over these spaces: uncertainty-driven methods ask *where the agent should sense or query next*; competence-driven methods ask *which embodied decision or action frontier should be practiced next*; and reachability-driven methods ask *how exploration can preserve a diverse repertoire of behaviours and keep future learning options open*.

Uncertainty-Driven. To reduce embodied uncertainty, an agent must decide what to observe before it can act reliably. Partial observability is not merely a passive data limitation: the agent can move its camera, body, microphone, or end-effector

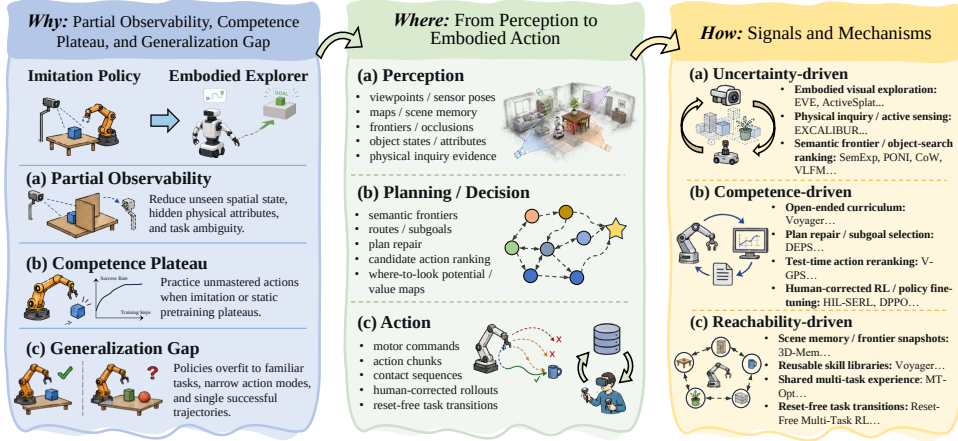


Fig. 5: Why-Where-How overview of Level 3 exploration in embodied agents. **Why:** partial observability, competence plateaus, and physical cost motivate exploration for uncovering hidden states, expanding skills, and preserving recoverable options. **Where:** perception, planning, and action define the embodied loop in which exploration is instantiated. **How:** uncertainty-driven, competence-driven, and reachability-driven mechanisms operate across these embodied locations.

to turn hidden states into evidence. Thus, uncertainty-driven embodied exploration operates over *perceptual actions*, such as viewpoints, sensor poses, path segments, visual crops, or physical probes. This is the Level3 form of Criterion1: information gain must be grounded in situated sensing.

Embodied visual exploration provides the basic vocabulary for this problem. Ramakrishnan et al. [30] show that exploration quality cannot be reduced to area coverage alone, since different policies may improve mapping, object discovery, localization, reconstruction, or downstream navigation. In this sense, embodied information gain may concern free space, object presence, distinctive views, unseen regions, or physical properties relevant to action.

Recent work makes this grounding explicit. ActiveSplat treats online 3D reconstruction as closed-loop view selection under a time budget [33]. 3D-Mem stores explored regions as memory snapshots and keeps unexplored regions as frontier snapshots for future movement and reasoning [34]. EXCALIBUR extends exploration from visual viewpoints to physical inquiry, showing that properties such as weight or material may require interaction before they can support language-grounded reasoning [35]. These methods shift embodied exploration from passive visual grounding to active perceptual, mnemonic, and experimental control.

Uncertainty-driven exploration also appears in planning and decision space. SemExp, PONI, CoW, and VLFM explore semantic maps, potential functions over frontiers, open-vocabulary object cues, and language-grounded frontier values to

decide where to search next [39, 56–58]. The key point is that embodied information gain is action-relevant: a viewpoint, frontier, camera pose, or object interaction is useful only insofar as it changes the robot’s next executable decision.

Competence-Driven. Expanding the embodied action frontier requires exploration when perception is adequate but the current policy cannot yet solve the task distribution. This is the central failure mode of passive behavior cloning and static VLA pretraining: the policy may imitate familiar demonstrations, yet fail under new object poses, long horizons, new embodiments, contact-rich dynamics, sparse rewards, or rare recovery states. The exploration target is therefore the agent’s capability boundary, not the environment map alone. This is the Level 3 form of Criterion 2: interaction should produce durable improvement in what the embodied policy can do.

In planning/decision space, competence-driven exploration concerns which subgoal, task, or action candidate should be attempted next. Voyager [40] makes this explicit through an automatic curriculum that maximizes open-ended exploration and stores acquired skills for later reuse. DEPS [41] targets the same problem from the plan-repair side: it describes execution feedback, explains failures, replans, and ranks candidate subgoals by estimated completion cost in Minecraft and other embodied domains. V-GPS [59] moves part of competence expansion to deployment time by reranking candidate robot actions with a learned value function before execution. In all three cases, the decision is tested through embodied interaction and downstream task completion.

In action space, competence-driven exploration targets the embodied action policy itself. HIL-SERL [44] interleaves autonomous exploration with human corrections, allowing the robot to learn from its mistakes during difficult dexterous tasks. DPPO [45] uses policy-gradient fine-tuning to produce structured, on-manifold exploration for diffusion policies, including sparse-reward robot settings. The explored objects are action chunks, contact sequences, and recovery trajectories that static demonstrations did not cover.

Reachability-Driven. Maintaining a diverse and accessible repertoire of behaviours is especially critical in embodied agents, where the training distribution can easily collapse onto a narrow set of familiar manipulation trajectories, foreclosing alternative action modes before they can be evaluated. The agent must therefore keep a broad set of task-relevant states, reset strategies, and reusable skills reachable throughout learning. This is the Level 3 form of Criterion 3: exploration should preserve future learning options by sustaining multi-task coverage and cross-task skill transferability.

In embodied learning, reachability asks whether exploration preserves access to a diverse set of feasible action modes, reset strategies, and reusable skills rather than collapsing onto one familiar manipulation trajectory. MT-Opt [46] operationalizes this idea by sharing exploration, experience, and representations across real-world tasks, while Reset-Free Multi-Task RL [47] shows that a repertoire of tasks can make other tasks’ start states reachable without manual resets.

Level 3 embodied exploration is thus best understood as controlled interaction with the world. The agent explores perceptual viewpoints to reduce uncertainty, practices action frontiers to expand competence, and constrains or reshapes the search so that

Table 3: Level 3 Embodied Agent Exploration Organized by Where and How

Where \ How	Uncertainty-driven	Competence-driven	Reachability-driven
Perception	Select informative views, sensor poses, physical inquiries, or reconstruction paths to expose hidden spatial and object properties: EVE [30], EXCALIBUR [35], ActiveSplat [33].	–	Store explored regions and keep frontier regions actionable through scene memory: 3D-Mem [34].
Planning / decision	Rank semantic frontiers, potential maps, object-localization cues, or VLM-valued frontiers before movement: SemExp [56], PONI [57], CoW [39], VLFM [58].	Select curriculum goals, candidate subgoals, or plan repairs that target unmastered embodied tasks: Voyager [40], DEPS [41].	Reuse skill libraries, keep fallback subgoals executable, or screen candidate actions before execution: Voyager [40], DEPS [41], V-GPS [59].
Action	–	Use corrective demonstrations, online RL, or diffusion-policy fine-tuning to practice action chunks and contact-rich trajectories: HIL-SERL [44], DPPO [45].	Share behaviour repertoires across tasks and use tasks as resets for one another: MT-Opt [46], Reset-Free Multi-Task RL [47].

physical exploration remains recoverable while keeping future skill transfer possible. This is qualitatively different from Level 4: the central object here is still the embodied interaction loop, not a learned simulator as an independent imagination space.

3.3 Level 4: Prospector – Imagination-Space Exploration

At Levels 1–3, exploration operates within spaces the agent directly inhabits. Level 4 introduces a qualitatively different regime: exploration within learned world models—internal simulators that approximate environment dynamics and enable imagination-driven planning [60–62]. This shift transforms exploration from costly physical trial-and-error to efficient mental simulation, where agents evaluate counterfactual futures and discover strategies without real-world consequences.

The world model serves as a cognitive sandbox where the agent navigates latent manifolds, generates simulated rollouts, and probes epistemic boundaries [63] (Figure 6, Middle). However, this dual exploration regime introduces unique challenges: prediction errors compound over imagined horizons, uncertainty must be disentangled from irreducible noise, and safety-critical details risk being lost in latent compression. The exploration objective shifts from “which action to take” to “which latent configurations are worth visiting, which imagined trajectories are physically plausible, and how can intrinsic signals guide exploration when external feedback is absent.” Table 4 provides a comprehensive taxonomy of nineteen core methods for exploration within world models; six additional infrastructure papers are discussed in the body text.

Table 4: Where–How Mapping of Level 4: World-Model Exploration Methods (19 Key Papers)

Where / How	Curiosity & Uncertainty-driven	Competence & Optimistic Planning	Structured, Semantic & Self-supervised
Imagined Space	Plan to explore via ensemble disagreement in imagined states: Plan2Explore [64]; maximize Bayesian novelty with forward-model ensembles: MAX [65];	Archive and return to promising states: Go-Explore [66]; RBMLE-based optimistic dynamics: Optimistic WM [67].	–
Risk Frontier	–	Bayesian dynamics + risk-seeking iLQR to resolve model uncertainty at operational boundaries: Curious iLQR [68].	–
Value Landscape	Prediction error as intrinsic reward: Schmidhuber [69]; variational information maximization: VIME [70]; pure curiosity generates diverse behaviors: SelMo [71]; retroactive curiosity avoids noise-hijacking: Hindsight [72].	–	Curiosity and competence co-evolve with WM learning: Mantiuk [73]; foundation models guide semantic exploration: SENSEI [74].
Latent Manifold	Self-supervised temporal representations for novelty detection: LWM [75]; geometry of WM (Euclidean vs. projective) affects epistemic value: Sergeant-Perthuis [76].	Ensemble disagreement as curiosity signal in Dreamer framework: InDRiVE [77].	Relational inductive biases for compositional object exploration: Sancaktar [78]; object-centric abstraction with discriminative WM: GX-Chen [79]; exploration from self-supervised WM training alone: Santana [80]; infer symbolic WMs from unguided exploration: One Life [81]; separate task-relevant and distractor information: SeeX [82].

3.3.1 Why: The Dual Exploration Problem

The transition to Level 4 addresses a fundamental tension: real-world experience is necessary but physically expensive. This creates the *Dual Exploration Problem*: agents must simultaneously gather physical data to refine world-model fidelity *and* search imagined trajectories to extract policies [64, 65].

Compounding Errors and Reality Drift. World models are recursive self-simulators where infinitesimal errors compound exponentially [63]. Without accurate imagined rollouts, exploration becomes misleading—agents optimize for fictional states that diverge from physical reality. Non-reconstruction world models such as R2-Dreamer address this by removing reconstruction losses [83], producing more robust trajectories for exploration.

The Noise-Hijacking Trap. Active exploration within world models is easily hijacked by stochastic distractions. Curiosity in Hindsight applies curiosity retroactively to avoid noise-hijacking in stochastic environments [72]. Optimistic World

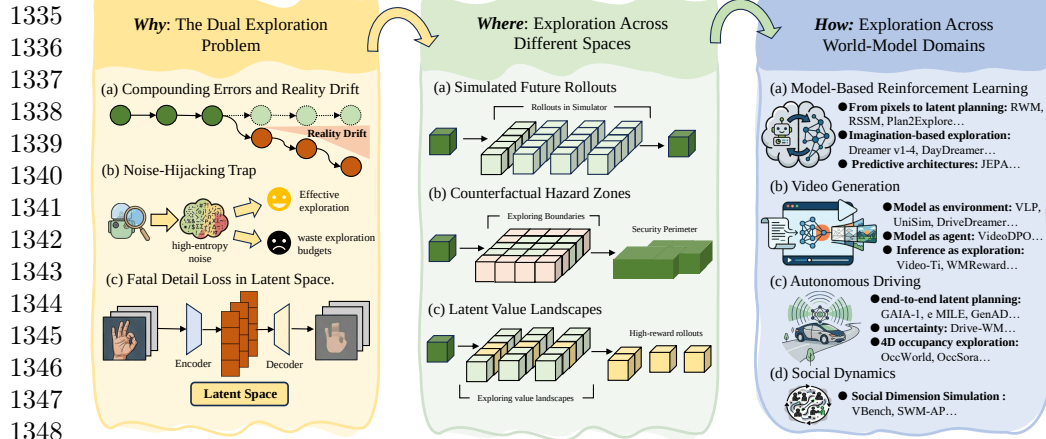


Fig. 6: Overview of the three core dimensions of exploration in world models. (Left) Motivation (Why): Addressing the dual exploration problem by mitigating reality drift, noise-hijacking traps, and fatal detail loss in latent spaces. (Middle) Exploration Space (Where): Expanding trial-and-error boundaries through simulated future rollouts, counterfactual hazard zones for safety, and latent value landscapes for high rewards. (Right) Application Domains (How): Leveraging these mechanisms across model-based RL, video generation, autonomous driving, and social dynamics to enable imagination-based exploration and end-to-end latent planning in complex systems.

Models incorporate RBMLE-based optimism directly into model learning, biasing transitions toward high-reward outcomes without explicit uncertainty estimates [67]. The SeeX framework separates endogenous and exogenous information within the world model, ensuring exploration focuses on task-relevant dynamics rather than visual distractors [82].

Fatal Detail Loss in Latent Space. World models compress high-dimensional streams into low-dimensional latents, discarding safety-critical details [60, 61]. When critical details are lost, exploration operates on impoverished representations that miss physical constraints essential for safe behavior. DayDreamer demonstrated that careful latent-space design enables successful transfer from imagination to physical robots [84].

3.3.2 Where: Exploration Across Different Spaces

Imagined Space. Action-space exploration within world models involves generating imagined trajectories to discover effective behaviors [65]. Go-Explore achieved breakthrough results by combining world models with an explicit state archive, returning to promising states before further exploration [66]. DreamerV3 demonstrated that imagined rollouts enable diamond acquisition in Minecraft through extensive latent-space search [63]. The fundamental advantage is *computational parallelism*: imagined trajectories are generated thousands of times faster than physical experiments.

Risk Frontier. Safety-critical exploration requires probing operational boundaries to identify failure modes [68]. Curious iLQR combines Bayesian dynamics modeling with risk-seeking control, explicitly resolving model uncertainty during trajectory optimization [68]. Without deliberate adversarial exploration, agents remain blind to failure modes at the boundaries of their competence.

Value Landscape. In the absence of dense external rewards, agents construct internal value landscapes through intrinsic motivation. Schmidhuber’s foundational work proposed using prediction error as intrinsic reward in model-building controllers [69]. VIME formalized exploration as variational information maximization about environment dynamics [70]. SENSEI leverages foundation models to guide *semantic* exploration, learning versatile world models through meaningful behavior rather than random wandering [74]. Mantiuk et al. demonstrated that curiosity and competence co-evolve with world model learning during exploration, creating a feedback loop that drives autonomous skill acquisition [73].

Latent Manifold. The compressed latent spaces learned by world models form manifolds where semantically similar states cluster [60, 61]. One Life to Learn demonstrates that symbolic world models can be inferred from minimal unguided exploration in stochastic environments [81]. Santana et al. show that exploration policies can be learned entirely from self-supervised world model training without external rewards [80]. Sergeant-Perthuis et al. revealed that the *geometric structure* of the world model (Euclidean vs. projective) fundamentally impacts epistemic value and curiosity-driven behavior [76].

3.3.3 How: Exploration via World-Model Paradigms

Model-Based Reinforcement Learning

Intrinsic motivation within world models. The foundational mechanism for exploration in world models is *intrinsic motivation*—self-generated reward signals that drive agents toward novel or informative states within the learned latent space [69]. DeepMind’s SelMo demonstrated that pure curiosity optimization within a world model generates meaningful, diverse behaviors in manipulation and locomotion without any task-specific rewards [71]. However, prediction-error-based curiosity suffers from the noisy-TV problem in stochastic environments. Jarrett et al. addressed this through Curiosity in Hindsight, which applies curiosity retroactively to avoid noise-hijacking [72]. VIME offers a principled implementation through variational information maximization [70].

Ermolov and Sebe introduced Latent World Models (LWM), which learn self-supervised temporal representations and use them within a world model for novelty detection in hard-exploration Atari games [75]. LWM explicitly arranges embeddings respecting temporal distance, making the learned representation robust to stochasticity and suitable for exploration. Sergeant-Perthuis et al. took a fundamentally different approach, analyzing how the *geometry* of the world model’s internal representation space (Euclidean vs. projective group structure) impacts epistemic value and curiosity-driven exploration [76]. They formally demonstrated that projective geometry induces nonlinear magnification of information, generating approach behaviors toward objects of interest, while Euclidean geometry has no such effect.

1427 **Active exploration via model planning.** The Dreamer lineage demon-
 1428 strates progressively sophisticated exploration within learned latent spaces [60–63].
 1429 Plan2Explore achieves task-agnostic exploration by maximizing latent disagreement
 1430 among ensemble dynamics models within the imagined space [64]. MAX (Model-
 1431 Based Active Exploration) uses an ensemble of forward models to plan observing
 1432 novel events, optimizing behavior with respect to Bayesian novelty measures [65]. The
 1433 latest Optimistic World Models incorporate RBMLE-based optimism directly into
 1434 model learning, representing a paradigm shift from UCB-style exploration to fully
 1435 gradient-based optimistic dynamics [67].

1436 **Structured and curiosity-driven exploration.** Moving beyond unstructured
 1437 state-space coverage, recent advances inject structure into world-model exploration.
 1438 Sancaktar et al. used relational inductive biases in structured world models for compo-
 1439 sitional multi-object environments, achieving zero-shot generalization to downstream
 1440 manipulation tasks [78]. GX-Chen et al. proposed a discriminative world model with
 1441 object-centric abstraction, where items are modeled at a higher state abstraction level
 1442 and attribute changes at a higher temporal abstraction level, enabling efficient count-
 1443 based exploration planning [79]. Bechtle et al. combined Bayesian dynamics modeling
 1444 with risk-seeking iLQR control, explicitly resolving model uncertainty during tra-
 1445 jectory optimization [68]. Mantiuk et al. revealed the two-way interaction between
 1446 curiosity and competence during world model learning, showing how exploratory
 1447 behavior and representational quality co-evolve [73].

1448

1449 *Video Generation*

1450 **Non-reconstruction world models for exploration.** A key challenge in using
 1451 world models for exploration is the computational overhead and representational brit-
 1452 tleness introduced by reconstruction objectives. R2-Dreamer eliminates both decoders
 1453 and data augmentation, demonstrating that redundancy reduction paradoxically
 1454 improves exploration by forcing the model to capture only task-relevant features [83].
 1455 These architectures enable faster imagined rollouts and reduce the compounding errors
 1456 that plague reconstruction-based world models during long-horizon exploration.

1457 **Visual distractor separation for robust exploration.** Real-world observa-
 1458 tions often contain task-irrelevant visual distractors that mislead intrinsic reward
 1459 signals. Huang et al. proposed SeeX, a bi-level optimization framework that trains a
 1460 *separated* world model to extract endogenous (task-relevant) and exogenous (distrac-
 1461 tor) information [82]. The inner optimization minimizes uncertainty in the endogenous
 1462 state space, while the outer optimization learns a policy on imaginary trajectories to
 1463 maximize task-relevant uncertainty. This ensures that exploration is directed toward
 1464 genuinely informative state transitions rather than visual noise.

1465

1466 *Autonomous Driving*

1467 **Disagreement-based vehicle exploration.** Autonomous driving presents unique
 1468 challenges for world-model exploration: safety constraints are paramount, and the
 1469 environment is highly stochastic with complex multi-agent interactions. InDRiVE
 1470 addresses these challenges by using intrinsic disagreement among ensemble dynam-
 1471 ics models as a curiosity signal within a Dreamer framework, driving exploration
 1472

in vehicle domains while maintaining robustness to stochastic traffic dynamics [77]. This approach ensures that exploration is directed toward regions of epistemic uncertainty—such as rare traffic configurations or novel road geometries—rather than unlearnable noise, enabling safe and efficient policy learning for autonomous navigation.

Social Dynamics

Competence-curiosity co-evolution. The interplay between curiosity and competence (empowerment) shapes how world models evolve during exploration in social and multi-agent environments. Mantiuk et al. demonstrated that world model learning and exploratory behavior form a feedback loop: better world models enable more targeted curiosity-driven exploration, which in turn generates richer data that improves the world model [73]. This co-evolution mirrors social learning dynamics, where agents must balance exploring novel social strategies with leveraging accumulated competence.

Foundation-guided semantic exploration. SENSEI leverages foundation models to guide *semantic* exploration, ensuring that world model learning in complex social environments is driven by meaningful behavior rather than random state visitation [74]. By using semantic priors to guide exploration, agents can efficiently discover relevant social dynamics and normative structures, enabling rapid adaptation to new social contexts. One Life to Learn showed that even minimal unguided exploration can yield symbolic, interpretable world models, bridging the gap between low-level sensory exploration and high-level cognitive representations [81].

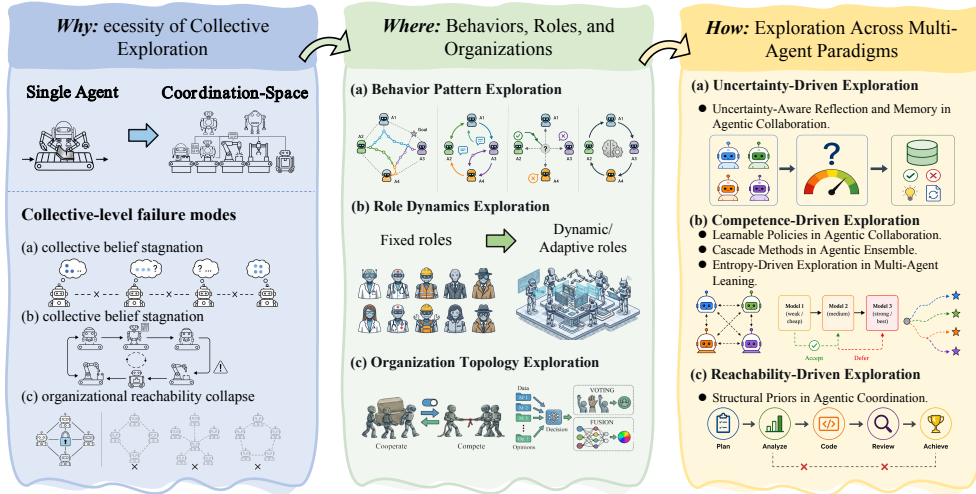


Fig. 7: The exploration framework diagram of coordination-space, from the three dimensions of "Why", "Where" and "How", explains the mechanism of exploring coordination-space.

1519 3.4 Level 5: Ecosystem — Coordination-Space Exploration

1520 At Level 5, the object of exploration becomes the coordination structure itself: how
1521 agents communicate, divide labour, assign roles, share memory, and adapt their inter-
1522 action topology. Level 5 thus introduces *coordination-space exploration*, where the
1523 central question is not only how one agent should explore, but how a population of
1524 agents should organise joint exploration to form adaptive, self-organising collective
1525 intelligence.

1527 3.4.1 Why: The Structural Necessity of Collective Exploration

1528
1529 At Level 5, exploration is necessary because collective intelligence does not emerge
1530 simply by adding more agents. A population may still exchange redundant informa-
1531 tion, follow brittle interaction routines, or lock agents into roles and topologies that
1532 no longer fit the task. The bottleneck therefore shifts from individual capability to
1533 collective organisation: agents must explore how to communicate, divide labour, verify
1534 one another, and adapt their coordination structure.

1535 This necessity appears in three collective-level failure modes. First, **collective**
1536 **belief stagnation** occurs when agents hold partial but disconnected observations,
1537 preventing the system from forming a coherent shared understanding. Second, **coor-**
1538 **dination value stagnation** occurs when capable agents fail to improve collective
1539 performance because the workflow, role assignment, or model-routing strategy is mis-
1540 matched to the task. Third, **organizational reachability collapse** occurs when
1541 the system prematurely commits to a fixed topology or protocol, making alternative
1542 coordination structures effectively unreachable under changing conditions.

1543 These failures show that Level 5 exploration must target the coordination space
1544 itself. The object of exploration is no longer an individual trajectory, but the collective
1545 structure through which agents interact. Accordingly, coordination-space exploration
1546 unfolds across *behavior patterns*, *role dynamics*, and *organization topology*, enabling
1547 heterogeneous agents to move from parallel individual reasoning toward adaptive,
1548 self-organising collective intelligence.

1549

1550 3.4.2 Where: Behaviors, Roles, and Organizations

1551
1552 Exploration at Level 5 unfolds in the interaction space where agents exchange informa-
1553 tion, organise cooperation, distribute responsibilities, and adapt their relations over
1554 time. The key question is therefore not only which agents are available, but where
1555 exploration occurs within the collective: in communication channels, collaboration
1556 protocols, and role configurations.

1557 **Behavior Pattern Exploration.** Behavior-pattern exploration concerns the col-
1558 lective behaviours through which agents act, react, and adapt during interaction.
1559 In multi-agent systems, the object of exploration is not merely the action of an
1560 individual agent, but the joint trajectory, peer response, disagreement pattern, and
1561 iterative reasoning routine that emerge from agent-agent interaction. This issue
1562 is especially salient in MARL, where the joint action space grows combinatorially
1563 and each agent’s learning changes the effective environment faced by others. As
1564 a result, exploration cannot be reduced to independent trial-and-error by isolated

Table 5: Summary of Level 5: Coordination-Space Exploration Methods

Where / How	Uncertainty-driven	Competence-driven	Reachability-driven
Behavior pattern	Trajectory-level entropy TEE [85], EMAX [86], joint intrinsic-motivation methods [87], peer-incentivized exploration [88], and behavioural-novelty-based exploration [89].	Feedback-driven collaboration tuning and weak-to-strong workflow optimisation: CMAT [90] and W4S [91].	-
Role dynamics	Active-Inference Orchestrator [92] and AgentVerse [93].	Role competence and team composition evolve: CMAT [90] and AgentVerse [93]. Weak-to-strong model cascade: AutoMix [94], Cascade Routing [95].	Keep multi-agent problem solving within reachable collaboration patterns: CAMEL [96] and MetaGPT [97].
Organization topology	Adjusts the interaction structure under unresolved uncertainty: Active-Inference Orchestrator [92].	Learnable coordination policies: CMAT [90] and W4S [91].	Constrains or evolves the collaboration graph: MetaGPT [97], MACNET [98], and Evolving Orchestration [99].

agents; it must discover collective behaviours that remain reachable only through coordinated interaction. Existing MARL studies therefore explore behaviour patterns through trajectory-level diversity, coordinated novelty, peer influence, and meaningful behavioural deviation. For example, trajectory-entropy methods encourage agents to maintain diverse collective trajectories, while joint intrinsic-motivation methods reward coordinated novelty that cannot be identified from individual novelty alone [85, 87]. Similarly, peer-incentivized and behavioural-novelty-based approaches transform inter-agent influence into learning signals, allowing the system to expand what it can reliably coordinate and accomplish [88, 89]. In this sense, behaviour-pattern exploration identifies where self-evolution first appears: in the repeated interaction routines through which agents gradually reshape their collective behavioural repertoire.

Role Dynamics Exploration. (1) In Agentic Collaboration, role dynamics determine how responsibilities are allocated, stabilised, and revised during collective problem solving. Some systems rely on stable division of labour to reduce coordination ambiguity. CAMEL[96] is a role-playing multi-agent collaboration framework. Other systems preserve a fixed role schema while improving the competence of each role over time. CMAT, for example, retains stable user–assistant–checker roles while strengthening role competence through feedback-driven tuning and memory [90]. A more flexible alternative is dynamic team formation: AgentVerse enables agent recruitment and role adjustment, allowing team composition to change when bottlenecks emerge [93]. Role exploration therefore spans a spectrum from stable specialisation to adaptive reconfiguration, with the central trade-off lying between coordination reliability and task-specific flexibility; (2) In Agentic Ensemble, the system explores when to halt computation along a weak-to-strong model cascade and which model should ultimately serve as the decision maker [94].

Organization Topology Exploration. Evolving Orchestration [99] focuses on how agents are dynamically scheduled. Traditional multi-agent systems typically adopt fixed architectures, such as predefined roles, fixed execution sequences, and static communication protocols. In real-world tasks, however, different stages demand the participation of distinct agents. Evolving Orchestration employs a central orchestrator analogous to a "controller". Based on the real-time task state, this orchestrator dynamically determines which agent to activate, the timing of its execution, and whether additional agents need to be invoked. As a result, the collaboration structure transforms from a static pipeline into a learnable, evolvable process.

1620

1621 **3.4.3 How: Exploration Across Multi-Agent Paradigms**

This section addresses how collective exploration is initiated, guided, and sustained across four distinct multi-agent paradigms. Within each paradigm, we identify how the three methodologies—uncertainty-driven, competence-driven, and reachability-driven—manifest in practice.

1626

1627 *Uncertainty-Driven Exploration*

1628 **Uncertainty-Aware Reflection and Memory in Agentic Collaboration.**

Uncertainty-aware orchestration frames coordination as adaptive control over interaction and experience. Active-inference-based orchestration uses variational free energy to balance expected information gain against execution cost, enabling the system to replan, backtrack, or suppress redundant interactions when uncertainty remains high [92]. Beyond immediate routing, persistent memory further extends orchestration to long-horizon experience management: agents must decide which successful executions, failures, reflections, and reusable procedures should be stored, compressed, and retrieved to shape future collaboration. AGENTVERSE regards multi-agent collaboration as an iterative team problem-solving process[93].

1639

1640 *Competence-Driven Exploration*

1641 **Learnable Policies in Agentic Collaboration.**

Learnable orchestration treats coordination itself as an optimisation problem rather than a fixed protocol. CMAT improves collaborative competence through feedback-driven tuning of agent roles and interactions, while W4S formulates workflow optimisation as a multi-round decision process guided by weak-to-strong supervision [90, 91]. These approaches shift orchestration from static protocol design to adaptive policy learning, allowing the collective to allocate computation according to task demands, intermediate outcomes, and observed coordination failures.

1649 **Cascade Methods in Agentic Ensemble.**

Cascade methods in the field of Agentic Ensemble arrange multiple LLMs in a sequence, usually from cheaper or weaker models to stronger ones, and decide whether the current answer is good enough or should be deferred to the next model. Unsupervised methods make this decision with simple signals such as user judgment, answer consistency, or output uncertainty/confidence [94]. Supervised methods learn the deferral rule from data, using post-hoc deferral, scoring functions, or MDP-based routing to balance answer quality and inference cost [95].

1656

Entropy-Driven Exploration in Multi-Agent Learning. Entropy-driven exploration sustains collective exploration by preventing agents from prematurely collapsing into a single behavioural or coordination mode. Unlike independent action noise, which treats each agent’s exploration separately, entropy-driven methods operate at the level of collective behaviour: they preserve diverse trajectories, latent coordination modes, or disagreement patterns so that the system can continue to access alternative joint behaviours. MAVEN introduces latent variables to induce temporally consistent diverse behaviours, thereby maintaining structured exploration in cooperative MARL [100]. TEE further shifts the emphasis from action-level randomness to trajectory-level diversity by maximising trajectory entropy in a contrastive representation space [85]. EMAX uses ensemble value disagreement to identify uncertain interaction regions and guide agents toward areas where collective knowledge remains incomplete [86]. These methods instantiate uncertainty-driven exploration because they preserve multiple feasible coordination patterns and reduce the risk that the multi-agent system converges too early to a locally stable but suboptimal equilibrium.

Reachability-Driven Exploration

Structural Priors in Agentic Coordination. Explicit structural priors make the coordination space tractable by constraining how agents are organised, sequenced, and allowed to exchange information. Role-specialised workflows decompose complex tasks into staged responsibilities, as exemplified by MetaGPT, whereas graph-structured systems such as MACNET organise agents through directed acyclic dependencies and transmit refined artifacts rather than unrestricted dialogue histories [97, 98]. These designs use topology, role assignment, and aggregation rules as reachability constraints: they suppress degenerate interaction patterns, reduce context expansion, and preserve access to productive coordination trajectories.

4 Evaluation of Agent Exploration

Evaluation is where epistemic exploration becomes falsifiable. It should not merely list agent benchmarks: it should specify what evidence separates exploration from ordinary task performance. In this review, an exploration claim is credible only when the evaluation can show that the system acquired missing information, used it to expand competence, and preserved access to non-redundant future options. The following section therefore asks three questions: what harness makes exploration observable, where such evaluation occurs across the five-level trajectory, and how the resulting evidence maps back to the three criteria introduced above.

4.1 A Minimal Harness for Exploration Evaluation

Most agent evaluations are built for exploitation: a task fixes a target, the model produces an answer, a grader checks correctness, and a binary outcome is recorded [101]. Exploration requires a different reading of the same harness. **Task** design should make success depend on missing information, alternative hypotheses, or feedback-sensitive revision. **Metrics** should compare one-shot competence with search-enabled performance, using signals such as $\Delta_{\text{explore}} = \text{pass}@k - \text{pass}@1$ while normalising

Table 6: Core References for Evaluating Exploration Across Levels 2–5

Level	Exploration Space	Evaluation Target	Core References
Level 2	Reasoning space	Does search expand reachable answers or task-relevant context beyond one-shot competence?	FrontierMath [103]; SWE-Explore [104]
Level 3	Interaction space	Does external action uncover hidden state and improve subsequent control?	AI Research Agents for ML [105]; Trainee-Bench [106]; Theory of Space [107]
Level 4	Imagination space	Does model-based or benchmark-level search remain grounded and improve real or held-out decisions?	WorldTest [108]; TRACE [109]
Level 5	Coordination space	Does collective search improve the shared information state beyond matched single-agent baselines?	MultiAgentBench [110]; TRACE [109]

for test-time compute [102]. **Grader** design should recognise useful discoveries and informative failures rather than only final correctness. **Outcome** should distinguish immediate success from boundary expansion and transfer. **Environment** should allow partial observability, feedback, backtracking, trial isolation, and recoverable errors; without these properties, the agent has little opportunity to explore.

The key diagnostic is counterfactual: what can the agent reach *with* search, interaction, imagination, or coordination that it could not reach *without* them? A larger trace, more samples, or more agents is not enough. The extra activity counts as exploration only if it changes the information state, improves later choices, or keeps distinct productive options reachable.

4.2 Where: Evaluation Across Levels 2–5

The five-level trajectory in Section 3 remains intact: Level 1 is the reactive baseline, while Levels 2–5 introduce explicit exploration spaces that require distinct evaluation evidence. Table 6 organises the core references by the location of exploration rather than by generic benchmark category.

Level 2: reasoning-space evaluation. Here exploration is internal search over reasoning paths, code context, or verifiable hypotheses. FrontierMath [103] tests capability-boundary reasoning where additional attempts may expose whether search

moves the model beyond one-shot competence, while SWE-Explore [104] shifts software evaluation from final patch correctness to the repository exploration that makes later repair possible. The main false positive is length: longer chains or wider sampling are exploratory only when they produce distinct, verifiable progress.

Level 3: interaction-space evaluation. Exploration now occurs through action in digital, research, workflow, or embodied environments. AI Research Agents for ML [105] evaluates search policies for research agents; Trainee-Bench [106] tests active exploration in dynamic workflows; Theory of Space [107] tests whether multimodal agents build and revise spatial beliefs from partial observations. These settings ask whether tool use, sensing, or workflow interaction actually reduces hidden uncertainty, rather than merely rewarding interface fluency.

Level 4: imagination-space evaluation. Prospector-level evaluation asks whether imagined exploration helps real decisions. WorldTest [108] separates reward-free exploration from later environment-level queries, testing whether the agent has learned a reusable world model. TRACE [109] gives a benchmark-level analogue: test-time exploration is useful only when evolved tasks remain valid, difficult, efficient, and supported by reproducible trajectories. The trap is model exploitation, where diverse imagined rollouts exploit a flawed simulator rather than improving grounded control.

Level 5: coordination-space evaluation. At the ecosystem level, the unit of evidence is collective information gain. MultiAgentBench [110] evaluates collaboration and competition among LLM agents, while TRACE [109] also supports benchmark-ecosystem evolution through validated trajectories. The right comparison is not a single agent versus many agents with more total budget, but a matched-budget baseline that tests whether communication, role specialisation, or task evolution reduces redundant search.

4.3 How: Mapping Evaluation to the Three Criteria

The harness and level-specific benchmarks above make Information Gain, Value Improvement, and Epistemic Reachability testable. **Information Gain (C1)** is evaluated when a benchmark can observe uncertainty reduction: repository facts discovered before repair in SWE-Explore [104], hidden workflow state uncovered through active exploration in Trainee-Bench [106], spatial beliefs revised from partial views in Theory of Space [107], and world-model structure recovered after reward-free exploration in WorldTest [108]. These benchmarks ask whether exploration changes what the agent knows, not just what it outputs.

Value Improvement (C2) is evaluated when the acquired information changes what the agent can do. FrontierMath and test-time compute scaling expose capability-boundary gains from additional search [102, 103]; SWE-Explore links repository discovery to later software repair [104]; AI Research Agents for ML tests whether search policies improve experimental progress in MLE-bench [105]; and WorldTest asks whether an explored world model improves held-out environment-level decisions [108]. C2 therefore requires a downstream outcome, not merely a richer trace.

Epistemic Reachability (C3) is evaluated when a benchmark tests whether useful alternatives remain accessible. TRACE checks that benchmark evolution through

test-time exploration remains valid and reproducible rather than drifting into unsupported task variants [109]; MultiAgentBench supports the collective analogue by asking whether coordination improves shared problem solving beyond redundant parallel effort [110]; and Theory of Space makes spatial belief revision measurable under partial observation [107]. C3 is the criterion that rules out redundant samples, simulator-exploiting paths, and fixed roles that look active but close off future discovery.

This mapping is the difference between exploration evaluation and ordinary evaluation. Ordinary evaluation can report that an agent scored higher; exploration evaluation must show why the gain came from acquiring and using new information. Without C1, the agent may simply exploit prior competence. Without C2, it may gather irrelevant facts. Without C3, it may win locally while closing off future discovery. A benchmark belongs in this section only insofar as it helps make one of these distinctions measurable.

5 Open Challenges and Conclusion

5.1 Open Challenges

Despite rapid progress across reasoning models, agentic systems, world models, and multi-agent coordination, epistemic exploration remains far from solved. We summarise the most critical open challenges below.

Level 2: Reasoner. Current reasoners still face several open challenges in exploration. First, when confronted with difficult problems, they often lack the ability to deliberately and actively search for new reasoning paths beyond familiar templates, and instead mainly sample or re-rank already accessible trajectories. Second, during test-time reasoning, it remains unclear when exploration should be triggered and at which level it should operate—over tokens, full reasoning sequences, or latent representations—making the use of exploration budget inefficient. Third, in multi-turn dialogue, reasoners still lack effective mechanisms for interactive exploration with users, such as asking clarifying questions, testing assumptions, incorporating user feedback, and progressively revising hypotheses to reach more reliable conclusions.

Level 3: Agent. At Level 3, agents still face several open challenges in exploration. First, they lack the ability to explore and create new tools: most current agents can select from a predefined tool set, but rarely discover, compose, or construct new tools when existing ones are insufficient. Second, agents remain weak at exploring unknown options in the real world, where useful affordances, hidden states, and alternative action possibilities may not be specified in advance. Third, in task-oriented interaction, agents still lack the ability to actively guide users through exploration, such as asking targeted questions, proposing alternative actions, testing uncertain assumptions, and helping users jointly discover better task solutions.

Level 4: Prospector. At Level 4, world models enable agents to explore imagined futures before acting, but several challenges remain. First, current world models still provide limited support for counterfactual exploration and causal verification of real-world actions: they can often predict plausible futures, but struggle to test whether an action would truly cause a desired outcome under interventions. Second, agents must

balance efficient imagination with physical fidelity, since compact latent rollouts make planning scalable but may discard safety-critical details such as contact dynamics, friction, deformation, and physical constraints. Third, exploration within imagination remains insufficient: world models often simulate variations around familiar states and actions, but struggle to actively explore unknown environments, unseen affordances, and novel action possibilities that could expand the agent’s future competence.

Level 5: Ecosystem. At Level 5, ecosystem-level exploration still faces several key challenges. First, agents must learn how to use collective intelligence for exploratory discovery, identifying new actions, strategies, and solution paths that are inaccessible to any single agent. Second, collective task execution requires adaptive exploration orchestration: the system must dynamically explore which agents to activate, how to assign roles, when to communicate, and how to reorganize collaboration as task demands change. Third, feedback must support exploration-driven credit assignment and self-evolution, enabling the ecosystem to identify which agents, roles, communication patterns, or coordination structures expanded the search space or improved performance, and to use this signal to guide future collective exploration.

5.2 Conclusion

Exploration is not a peripheral component of intelligence, but a core mechanism through which intelligent systems remain adaptive, open-ended, and capable of growth. In this survey, we framed progress toward AGI through *epistemic exploration*, defined by three criteria: *Information Gain*, *Value Improvement*, and *Epistemic Reachability*. These criteria explain how agents reduce uncertainty, convert new information into capability gains, and preserve access to future possibilities. We further organised existing work along a five-level trajectory from *Responder* to *Reasoner*, *Agent*, *Prospector*, and *Ecosystem*, where exploration expands from reasoning paths to external actions, imagined futures, and coordination structures. This perspective suggests that the path toward general intelligence requires not only better optimisation, but also deeper and more sustained exploration.

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Code availability. Project materials are available at github.com/banyikun/agent_exploration.

Competing interests. The authors declare no competing interests.

1887 Annotated Key References

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1889 Foundation of Exploration

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1891 • Piaget [1]. *Key contribution:* A foundational account of cognitive development,
1892 showing how children actively construct knowledge through sensorimotor inter-
1893 action with the world. This work motivates the view that exploration precedes
1894 explicit instruction and is central to the emergence of adaptive intelligence.

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1896 • OpenAI [5]. *Key contribution:* Provides a public capability framing that dis-
1897 tinguishes chatbots, reasoners, agents, innovators, and organizations. Although
1898 not peer-reviewed, it is useful as a contemporary reference point for comparing
1899 capability-level taxonomies of progress toward increasingly agentic systems.

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1901 • Hills et al. [6]. *Key contribution:* Synthesizes exploration across cognitive, ecolog-
1902 ical, and computational domains, framing exploration as a general mechanism for
1903 adaptive search under uncertainty. This work supports the survey’s broader view
1904 that exploration is not domain-specific but a cross-cutting principle of intelligence.

1905

1906 • Berlyne [7]. *Key contribution:* Established a classical psychological theory of
1907 curiosity and exploratory behaviour, identifying novelty, complexity, surprise,
1908 and incongruity as major drivers of information-seeking. This provides an early
1909 conceptual basis for epistemic forms of exploration.

1910

1911 • Kidd and Hayden [8]. *Key contribution:* Provides an influential psychological
1912 account of curiosity as information-seeking behaviour, emphasizing how agents
1913 select observations to improve their internal models. This directly motivates the
1914 information-gain component of epistemic exploration.

1915

1916 • Auer et al. [9]. *Key contribution:* Introduced the UCB principle for balancing
1917 exploration and exploitation in multi-armed bandits. It is a canonical example of
1918 optimism under uncertainty and forms a theoretical foundation for uncertainty-
1919 driven exploration.

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1921 • Osband et al. [10]. *Key contribution:* Developed posterior sampling for reinforce-
1922 ment learning, showing how Bayesian beliefs over environments can guide efficient
1923 exploration. This work is closely aligned with the belief-directed perspective
1924 underlying epistemic exploration.

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1926 • Pathak et al. [11]. *Key contribution:* Proposed intrinsic curiosity based on self-
1927 supervised prediction error, demonstrating how agents can explore sparse-reward
1928 environments without dense external supervision. It is a representative modern
1929 method for curiosity-driven exploration in deep reinforcement learning.

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1931 • Ban et al. [12]. *Key contribution:* Uses ensemble-based uncertainty to guide explo-
1932 ration, providing a practical mechanism for approximating epistemic uncertainty
through model disagreement. This connects directly to the uncertainty-driven
methodology discussed in the survey.

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Level 2: Reasoner — Reasoning-Space Exploration	1933
• Nguyen et al. [13] <i>Key contribution:</i> It shows that effective uncertainty sampling should prioritize epistemic uncertainty, rather than generic predictive uncertainty, because it better captures what the model can learn from new labels.	1934
• Yue et al. [14] <i>Key contribution:</i> It argues that RLVR mainly improves sampling efficiency by upweighting correct reasoning paths already present in the base model, rather than genuinely expanding the model’s underlying reasoning capacity.	1935
• Hou et al. [15] <i>Key contribution:</i> TreeRL introduces an on-policy tree-search-based RL framework that improves LLM reasoning by enabling more efficient reasoning-space exploration and dense process-level supervision without relying on a separately trained reward model.	1936
• Xing et al. [16] <i>Key contribution:</i> LATR (Lookahead Tree-Based Rollouts) improves trajectory-level exploration in RLVR by branching at high-uncertainty steps, using lookahead rollouts, and pruning similar trajectories to provide more informative reward signals for policy optimization.	1937
• Hu et al. [17] <i>Key contribution:</i> DIVER shows that global sequence-level diversity is a key driver of RLVR exploration, and introduces a diversity-based intrinsic reward with reward shaping to improve LLM reasoning performance and generalization.	1938
• Hao et al. [18] <i>Key contribution:</i> Coconut introduces continuous latent-space reasoning, where LLM hidden states are fed back as “continuous thoughts,” enabling more flexible multi-path reasoning than discrete CoT and reducing inference tokens.	1939
• Jia et al. [19] <i>Key contribution:</i> CoLA introduces a compact latent action space for LLM reinforcement learning, replacing token-level actions with higher-level semantic actions to improve controllability, exploration, and RL adaptation efficiency.	1940
• Zhang et al. [4] <i>Key contribution:</i> ReLaX proposes latent-dynamics-guided exploration for RLVR, using Koopman-based hidden-state analysis and Dynamic Spectral Dispersion to regulate exploration–exploitation and mitigate entropy collapse in large reasoning models.	1941
• Parashar et al. [3] <i>Key contribution:</i> E2H Reasoner shows that probabilistically scheduling RLVR training from easy to hard tasks, while gradually fading out easier tasks, enables LLMs to build reasoning skills more effectively than direct RL on hard tasks.	1942
• Wang et al. [20] <i>Key contribution:</i> AEPO shows that policy entropy is directly controllable during RL fine-tuning, and prevents entropy collapse by using temperature-adjusted REINFORCE updates to sustain exploration and improve LLM reasoning performance.	1943
• Wang et al. [21] <i>Key contribution:</i> ICPO introduces an intrinsic confidence-driven preference advantage that compares relative generation probabilities within response groups and combines it with verifiable rewards to reduce reward noise, prevent entropy collapse, and improve RLVR efficiency for LLM reasoning.	1944
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- 1979 • Xie et al. [22] *Key contribution:* UCAS introduces uncertainty-aware advantage
1980 shaping for RLVR, using response-level self-confidence and token-level certainty
1981 signals to refine credit assignment, sustain exploration, and mitigate entropy
1982 collapse in LLM reasoning.
- 1983 • Bae et al. [23] *Key contribution:* Online Difficulty Filtering shows that dynamically
1984 selecting medium-difficulty problems—where the current model has intermediate
1985 success rates—maximizes useful RLVR learning signals and improves reasoning
1986 efficiency.
- 1987 • Yang et al. [24] *Key contribution:* This paper shows that group-relative advantage
1988 estimation in RLVR is inherently biased—underestimating hard prompts and over-
1989 estimating easy ones—and proposes HA-DW to correct this imbalance for more
1990 effective reasoning RL.
- 1991 • Liang et al. [25] *Key contribution:* SvS shows that self-play problem augmen-
1992 tation—synthesizing answer-preserving variants from the model’s correct solu-
1993 tions—can sustain RLVR by maintaining policy entropy and improving Pass@k
1994 reasoning performance beyond standard Pass@1-focused training.
- 1995 • Jiang et al. [26] *Key contribution:* RLVRr extends RLVR from verifiable-answer
1996 tasks to open-ended generation by converting high-quality references into a verifi-
1997 able reward chain over content and style, providing more reliable RL signals while
1998 preserving diversity.
- 1999 • Wang et al. [2] *Key contribution:* It shows that a small set of high-entropy minority
2000 tokens act as key reasoning decision points, and selectively optimizing them can
2001 make RL for LLM reasoning more effective than updating all tokens uniformly.
- 2002

2003 **Level 3: Agent — Perception- and Action-Space Exploration**

2004 **Digital Agent**

- 2005 • Liu et al. [36] *Key contribution:* Agent0-VL introduces a self-evolving tool-
2006 integrated vision-language agent, using Solver–Verifier self-play to generate process
2007 rewards and repair signals for improving tool-use reasoning.
- 2008 • Hao et al. [50] *Key contribution:* EGPO improves robust function calling by adding
2009 entropy-aware exploration to RL, helping agents avoid premature collapse onto
2010 narrow tool-call patterns.
- 2011 • Zhang et al. [37] *Key contribution:* RAPO expands LLM-agent exploration by
2012 retrieving off-policy step-level traces during rollout, allowing the agent to explore
2013 beyond its own on-policy tool-use trajectories.
- 2014 • Guo et al. [51] *Key contribution:* E³-TIR improves tool-integrated reasoning by
2015 exploiting external or expert experience, providing useful tool-use branches that
2016 broaden the agent’s exploration space.
- 2017 • Qiao et al. [27] *Key contribution:* KnowSelf introduces agentic self-awareness,
2018 routing an agent among direct execution, reflection, and knowledge invocation
2019 according to self-perceived uncertainty.
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- Lu et al. [28] *Key contribution:* PilotRL trains LLM agents with a progressive curriculum from plan following to plan generation and planner–executor coordination, stabilizing long-horizon agentic exploration. 2025
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 - Hua et al. [48] *Key contribution:* EAPO proposes exploration-aware policy optimization, assigning fine-grained exploration rewards to intermediate actions to improve agentic credit assignment. 2028
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 - Feng et al. [38] *Key contribution:* GiGPO groups matched intermediate states across rollouts to compute state-level relative advantages, enabling finer-grained policy updates for LLM-agent training. 2031
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 - Zhao et al. [49] *Key contribution:* Absolute Zero demonstrates reinforced self-play reasoning without external data, showing how agents can generate their own task curriculum and learning signal. 2034
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 - Xu et al. [43] *Key contribution:* EPO regulates trajectory-level entropy in multi-turn LLM-agent RL, mitigating exploration–exploitation cascade failures and preserving behavioral diversity. 2038
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 - Li et al. [29] *Key contribution:* JitRL uses a memory repository of past state–action–return experiences to estimate just-in-time action advantages, enabling continual agent improvement without gradient updates. 2041
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 - Arumugam and Griffiths [31] *Key contribution:* PSRL Agent maintains a textual posterior over unknown environment dynamics and rewards, using posterior sampling to guide efficient exploration by LLM agents. 2044
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 - Li et al. [32] *Key contribution:* Search-o1 augments reasoning models with agentic search, retrieving and refining external knowledge to reduce uncertainty during long-form problem solving. 2047
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 - Wu et al. [42] *Key contribution:* ReSum-GRPO enables long-horizon search by compressing interaction history into reusable summaries, allowing agents to continue exploration beyond context-window limits. 2050
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- Embodied Agent** 2054
- Ramakrishnan et al. [30]. *Key contribution:* Defines embodied visual exploration and explains why coverage alone is not enough to evaluate active perception. We cite it as the baseline taxonomy for viewpoint selection, mapping, object discovery, and downstream embodied navigation. 2055
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 - Zhu et al. [35]. *Key contribution:* Shows that embodied agents must sometimes interact with objects to discover hidden physical properties such as weight, material, and relations. This supports the view that Level 3 exploration includes physical inquiry, not only passive sensing. 2059
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 - Li et al. [33]. *Key contribution:* Formulates active 3D reconstruction as closed-loop viewpoint and path selection under a limited budget. We cite it as a representative method for perception-space exploration in embodied agents. 2063
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 - Yang et al. [34]. *Key contribution:* Links embodied exploration to persistent 3D scene memory by distinguishing explored memory snapshots from unexplored frontier snapshots. This makes future evidence acquisition and spatial reachability explicit. 2066
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- 2071 • Chaplot et al. [56]. *Key contribution:* Provides a classical semantic exploration
2072 method for ObjectNav, using semantic maps to decide where the agent should
2073 search next. We cite it as an early anchor for planning-space exploration in
2074 embodied navigation.
- 2075 • Ramakrishnan et al. [57]. *Key contribution:* Makes the “where to look” deci-
2076 sion explicit through potential functions over object-goal frontiers. This supports
2077 the argument that embodied planning explores executable spatial options before
2078 acting.
- 2079 • Gadre et al. [39]. *Key contribution:* Combines open-vocabulary object localization
2080 with classical exploration for language-driven zero-shot ObjectNav. We cite it to
2081 show how language grounding can guide embodied search without task-specific
2082 training.
- 2083 • Yokoyama et al. [58]. *Key contribution:* Uses vision-language frontier maps to
2084 value candidate frontiers. It is cited as a concrete example of VLM-guided frontier
2085 selection for deciding where an embodied agent should explore next.
- 2086 • Wang et al. [40]. *Key contribution:* Demonstrates open-ended embodied explo-
2087 ration through automatic curriculum generation and a reusable skill library. We
2088 cite it as a competence-driven example where exploration expands what the agent
2089 can do over time.
- 2090 • Wang et al. [41]. *Key contribution:* Uses description, explanation, replanning, and
2091 subgoal selection to recover from failures in open-world embodied tasks. It sup-
2092 ports the claim that planning exploration includes revising and ranking executable
2093 subgoals before acting.
- 2094 • Nakamoto et al. [59]. *Key contribution:* Applies value guidance to rerank candidate
2095 actions of a generalist robot policy at deployment time. We cite it as an example
2096 of pre-execution exploration over feasible embodied actions.
- 2097 • Luo et al. [44]. *Key contribution:* Integrates human corrections with robot rein-
2098 forcement learning during difficult dexterous manipulation. This is cited because
2099 it turns costly physical mistakes into useful exploration data.
- 2100 • Ren et al. [45]. *Key contribution:* Fine-tunes diffusion policies with reinforce-
2101 ment learning and yields structured, on-manifold action exploration. We cite it to
2102 represent action-space exploration for modern robot policy classes.
- 2103 • Kalashnikov et al. [46]. *Key contribution:* Demonstrates large-scale multi-task
2104 robotic RL where tasks share exploration, experience, and representations. It
2105 supports the reachability-driven view that a broad task repertoire keeps future
2106 learning options open.
- 2107 • Gupta et al. [47]. *Key contribution:* Studies reset-free multi-task RL, where differ-
2108 ent tasks provide start states and recovery opportunities for one another. We cite it
2109 as a mechanism for maintaining embodied reachability without manual resetting.
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Level 4: Prospector – Imagination-Space Exploration	2117
• Schmidhuber [69]. <i>Key contribution:</i> Proposed using curiosity and boredom as intrinsic motivation signals, where the intrinsic reward is proportional to the learning progress of the agent’s internal world model. This establishes the theoretical basis for all prediction-error-based curiosity in world models.	2118 2119 2120 2121 2122
• Houthoofd et al. [70]. <i>Key contribution:</i> Formulated exploration as variational information maximization about environment dynamics, using a Bayesian neural network to derive information-gain intrinsic rewards. It provides a mathematically principled foundation for epistemic exploration in world models.	2123 2124 2125 2126
• Ermolov and Sebe [75]. <i>Key contribution:</i> Introduced Latent World Models (LWM), which learn self-supervised temporal representations for novelty detection in hard-exploration tasks. This demonstrates how self-supervised representation learning combined with world models enables efficient exploration.	2127 2128 2129 2130
• Groth et al. [71]. <i>Key contribution:</i> Demonstrated that purely curiosity-driven exploration within a world model can generate meaningful manipulation and locomotion behaviors without any task-specific rewards. It provides empirical validation that curiosity alone is sufficient for acquiring diverse motor skills.	2131 2132 2133 2134
• Jarrett et al. [72]. <i>Key contribution:</i> Addressed the noisy-TV problem through retrospective curiosity—asking “should I have predicted this?” after observing the next state. This reformulation makes curiosity signals invariant to stochastic transitions, enabling exploration in noisy environments.	2135 2136 2137 2138
• Sergeant-Perthuis et al. [76]. <i>Key contribution:</i> Showed that the geometric structure (Euclidean vs. projective) of the world model’s internal representation space fundamentally impacts curiosity-driven behavior. Projective geometry induces information magnification that generates approach behaviors toward objects of interest.	2139 2140 2141 2142 2143
• Shyam et al. [65]. <i>Key contribution:</i> Proposed Model-Based Active Exploration (MAX), which uses ensemble forward dynamics models to maximize Bayesian novelty. It establishes the paradigm of ensemble-disagreement-driven exploration within world models.	2144 2145 2146 2147
• Sekar et al. [64]. <i>Key contribution:</i> Introduced the first “plan to explore” method within self-supervised world models, decoupling exploration from task reward. This established the task-agnostic exploration paradigm via ensemble disagreement in imagined space.	2148 2149 2150 2151 2152
• Mete et al. [67]. <i>Key contribution:</i> Proposed Optimistic World Models (OWM), which incorporate RBMLE optimism directly into the world model learning objective. This eliminates expensive ensemble uncertainty estimation while achieving efficient exploration through fully gradient-based optimistic dynamics.	2153 2154 2155 2156
• Ecoffet et al. [66]. <i>Key contribution:</i> Introduced the “return-then-explore” paradigm using an explicit state archive and goal-conditioned return policies. It solved previously unsolved hard-exploration Atari games by proving that reliably returning to promising states is the key challenge.	2157 2158 2159 2160 2161 2162

- 2163 • Sancaktar et al. [78]. *Key contribution:* Combined curiosity-driven exploration
2164 with structured (object-centric) world models using relational inductive biases.
2165 This enables compositional exploration of object interactions and zero-shot
2166 generalization to novel manipulation tasks.
- 2167 • Chen et al. [79]. *Key contribution:* Proposed a discriminative world model with
2168 object-centric abstraction, enabling efficient count-based exploration planning at
2169 the object level. It provides a complementary discriminative approach to structured
2170 world-model exploration.
- 2171 • Bechtle et al. [68]. *Key contribution:* Combined Bayesian dynamics modeling with
2172 risk-seeking iLQR control to resolve model uncertainty during trajectory optimiza-
2173 tion. It integrates curiosity directly into model-based trajectory optimization for
2174 continuous control tasks.
- 2175 • Mantiuk et al. [73]. *Key contribution:* Analyzed the co-evolution of curiosity and
2176 competence during world model learning, revealing a feedback loop where better
2177 models enable more targeted exploration. It provides a principled understanding
2178 of the interplay between world model quality and exploration efficiency.
- 2179 • Santana et al. [80]. *Key contribution:* Showed that exploration policies can be
2180 learned entirely from self-supervised predictive world model training without any
2181 external reward. It proves that exploration can be fully self-sufficient through pure
2182 predictive learning.
- 2183 • Yu et al. [74]. *Key contribution:* Leveraged foundation models to guide semantic
2184 exploration, directing agents toward meaningful behaviors rather than random
2185 wandering. This represents a new paradigm where foundation models serve as
2186 “exploration mentors.”
- 2187 • Khan et al. [81]. *Key contribution:* Demonstrated that symbolic world models
2188 can be inferred from minimal unguided exploration in stochastic environments. It
2189 bridges low-level sensory exploration and high-level cognitive representations.
- 2190 • Huang et al. [82]. *Key contribution:* Proposed SeeX, a separated world model
2191 that extracts endogenous and exogenous information through bi-level optimiza-
2192 tion. This makes world-model exploration robust to visual distractors in complex
2193 environments.
- 2194 • Khanzada et al. [77]. *Key contribution:* Extended ensemble-disagreement curiosity
2195 to autonomous driving using a Dreamer-style world model. It demonstrates that
2196 world-model exploration can be adapted for safety-critical real-world domains.
- 2197 • Ha and Schmidhuber [60]. *Key contribution:* Introduced the foundational World
2198 Models architecture (VAE + RNN + Controller), establishing the paradigm of
2199 learned internal simulators for policy evolution. It is the architectural foundation
2200 for all subsequent world-model-based exploration methods.
- 2201 • Hafner et al. [61]. *Key contribution:* Introduced the Recurrent State-Space
2202 Model (RSSM) for latent-space planning from pixels. It provides the core world
2203 model architecture underlying the Dreamer lineage and latent imagination-based
2204 exploration.
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• Hafner et al. [62]. <i>Key contribution:</i> Introduced Dreamer, which trains actor-critic agents within the imagined latent space of an RSSM world model. It bridges world model architecture and practical model-based RL via imagined policy learning.	2209 2210 2211
• Hafner et al. [63]. <i>Key contribution:</i> Showed that a single world-model algorithm can master diverse domains without hyperparameter tuning. It demonstrates the scalability of world-model exploration and the challenges of compounding errors in long-horizon imagination.	2212 2213 2214 2215
• Morihira et al. [83]. <i>Key contribution:</i> Eliminated decoders and data augmentation from world models, showing that redundancy reduction improves exploration by capturing only task-relevant features. It addresses computational overhead and reality drift in long-horizon exploration.	2216 2217 2218 2219
• Wu et al. [84]. <i>Key contribution:</i> Transferred Dreamer from simulation to physical robots, demonstrating successful imagination-to-reality transfer. It shows that careful latent-space design enables world-model exploration in physical systems.	2220 2221 2222 2223
Level 5: Ecosystem — Coordination-Space Exploration	2224 2225
• Li and Zhu [85]. <i>Key contribution:</i> Proposes trajectory entropy maximization for efficient multi-agent exploration. The method shifts exploration from independent action noise to the diversity of joint trajectories, matching the Level 5 view that a collective system should preserve multiple coordination patterns before converging to a single behavioural routine.	2226 2227 2228 2229 2230
• Mahajan et al. [100]. <i>Key contribution:</i> Introduces latent-variable-based multi-agent variational exploration to induce temporally consistent diverse behaviours. It shows how structured latent modes can organize exploration over coordinated behaviour patterns, expanding the reachable set of team strategies beyond locally random or independently sampled actions.	2231 2232 2233 2234 2235
• Schäfer et al. [86]. <i>Key contribution:</i> Uses ensemble value-function disagreement to guide multi-agent exploration toward uncertain interaction regions. In coordination-space exploration, such disagreement identifies where the group has not yet formed reliable knowledge about joint value, making uncertainty itself a signal for deciding which collective behaviours to try next.	2236 2237 2238 2239 2240
• Toquebiau et al. [87]. <i>Key contribution:</i> Develops joint intrinsic motivation for coordinated exploration in multi-agent deep reinforcement learning. Its central idea is that novelty can be defined at the level of the group, rewarding coordinated behaviours that are informative only when agents act together rather than as isolated explorers.	2241 2242 2243 2244 2245
• Kölle et al. [88]. <i>Key contribution:</i> Proposes peer-incentivized multi-agent exploration, where agents encourage one another to visit novel or useful states. This turns peer influence into an exploration signal, connecting coordination-space search to the question of how agents can actively expand one another’s competence.	2246 2247 2248 2249 2250
• Pan et al. [89]. <i>Key contribution:</i> Frames curiosity-driven multi-agent exploration through contextual calibration and behavioural novelty. Its exploration signal emphasizes meaningful novelty in peer behaviour rather than arbitrary deviation,	2251 2252 2253 2254

2255 which aligns with Level 5 self-evolution: the group should discover interaction
2256 patterns that genuinely improve coordination.

- 2257 • Beckenbauer et al. [92]. *Key contribution:* Applies active inference to multi-agent
2258 orchestration in long-horizon tasks, balancing expected information gain and
2259 execution cost. This frames orchestration as uncertainty-aware control: the sys-
2260 tem can replan, backtrack, or suppress redundant interactions when the current
2261 coordination path is unlikely to produce useful information.
- 2262 • Chen et al. [93]. *Key contribution:* Presents AgentVerse as a framework for multi-
2263 agent collaboration and emergent behaviour, including agent recruitment and role
2264 adjustment. The framework connects role exploration with team formation: agents
2265 can be recruited, reassigned, or reorganized as task bottlenecks appear, rather than
2266 remaining locked into a fixed composition.
- 2267 • Liang et al. [90]. *Key contribution:* Proposes CMAT, a collaboration-tuning frame-
2268 work that improves small language models through feedback-driven multi-agent
2269 roles and memory. Its role schema remains relatively stable, but the competence
2270 of each role is improved through feedback and stored experience, illustrating
2271 competence-driven exploration inside a structured collaboration protocol.
- 2272 • Nie et al. [91]. *Key contribution:* Trains a weak meta-agent to coordinate stronger
2273 executor agents. The method treats workflow selection and coordination policy as
2274 learnable objects, so exploration occurs over which agents to activate, how to route
2275 tasks, and how to allocate computation across heterogeneous capabilities.
- 2276 • Dekoninck et al. [94]. *Key contribution:* Provides a unified view of routing and
2277 cascading for LLMs. In an agentic ensemble, routing and cascading instantiate
2278 exploration over model choice: the system tests whether a weaker or cheaper model
2279 is sufficient or whether the task should be escalated to a stronger model.
- 2280 • Aggarwal et al. [95]. *Key contribution:* Introduces AutoMix for automatically
2281 mixing language models through learned deferral and routing decisions. Its cas-
2282 cade mechanism links competence-driven exploration to practical inference: model
2283 selection becomes a search over answer quality, task difficulty, and computational
2284 cost.
- 2285 • Li et al. [96]. *Key contribution:* Introduces role-playing communicative agents for
2286 exploring LLM societies. Its explicit role assignments act as structural priors over
2287 coordination, reducing ambiguity in interaction and making stable multi-agent
2288 problem-solving trajectories easier to reach.
- 2289 • Hong et al. [97]. *Key contribution:* Designs MetaGPT as a multi-agent collab-
2290 orative framework with role-specialized workflows and structured artifacts. Its
2291 standardized handoffs and executable intermediate outputs illustrate how coordi-
2292 nation reachability can be improved by constraining collaboration into clear
2293 responsibility decompositions.
- 2294 • Qian et al. [98]. *Key contribution:* Proposes MACNET for scaling LLM-based
2295 multi-agent collaboration with graph-structured dependencies and artifact-based
2296 communication. The method makes organization topology and communication
2297 pathways explicit design objects, showing that exploration at Level 5 includes how
2298 information is routed, refined, and aggregated across agents.

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- Dang et al. [99]. *Key contribution:* Studies multi-agent collaboration via evolving orchestration, dynamically scheduling which agents participate and when. The work makes the collaboration structure itself evolvable, turning a static pipeline into an adaptive coordination process that can explore different activation orders, agent subsets, and workflow structures.

Evaluation of Agent Exploration

- Zhang et al. [104]. *Key contribution:* SWE-Explore directly evaluates how coding agents explore software repositories before producing a repair. We cite it because it separates repository context discovery from final patch correctness, making repository-level agent exploration measurable as information acquisition that later supports downstream software engineering performance.
- Toledo et al. [105]. *Key contribution:* This work frames AI research agents as search systems and compares different exploration policies in MLE-bench. It is cited as a representative evaluation of whether agentic search strategies actually improve experimental progress, rather than merely generating more trials.
- Fu et al. [106]. *Key contribution:* Trainee-Bench evaluates agents on active exploration in dynamic workplace workflows. We cite it because it tests whether an agent can uncover hidden workflow state, adapt to task streams, and convert exploration into usable performance in realistic digital environments.
- Zhang et al. [107]. *Key contribution:* Theory of Space evaluates how multi-modal agents know, see, and recall spatial environments under partial observation. It is cited because it makes spatial belief construction and revision measurable, providing an embodied evaluation case for information gain and epistemic reachability.
- Warrier et al. [108]. *Key contribution:* WorldTest evaluates whether reward-free exploration produces reusable world-model knowledge, using environment-level queries after interaction. We cite it as a central Level 4 example of testing whether imagined or model-based exploration is grounded in transferable environmental structure.
- Guo et al. [109]. *Key contribution:* TRACE studies benchmark-level test-time exploration by requiring evolved agent trajectories and task variants to remain valid, difficult, efficient, and reproducible. It is cited because it evaluates exploration not only by new task generation, but by whether the resulting benchmark evidence remains verifiable.
- Zhu et al. [110]. *Key contribution:* MultiAgentBench evaluates collaboration and competition among LLM agents. We cite it as the Level 5 evaluation anchor: collective exploration should be judged by whether communication and role specialization improve the shared information state beyond redundant parallel single-agent effort.

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